

The Solvelt Notebooks for Rust — an encyclopedia

128 Jupyter notebooks, every one project-authored, every one executed for real against the pure-Rust SUNDIALS 7.8.0 engine on this machine (128/128 green), organized into seven major topics. Each notebook lives in its own tagged subfolder together with its companion player script (`player_<tag>.py`), and each carries a prepended header cell with the exact commands to execute it, display its browser GUI, create/access its movies, and access its data — plus the full player script inline.

Every simulation below was integrated by the vendored pure-Rust port of SUNDIALS 7.8.0 CVODE (BDF + Newton + dense solver) — no C, no FFI, no unsafe code — and every notebook checks its own physics against closed forms before it is allowed to say ok.

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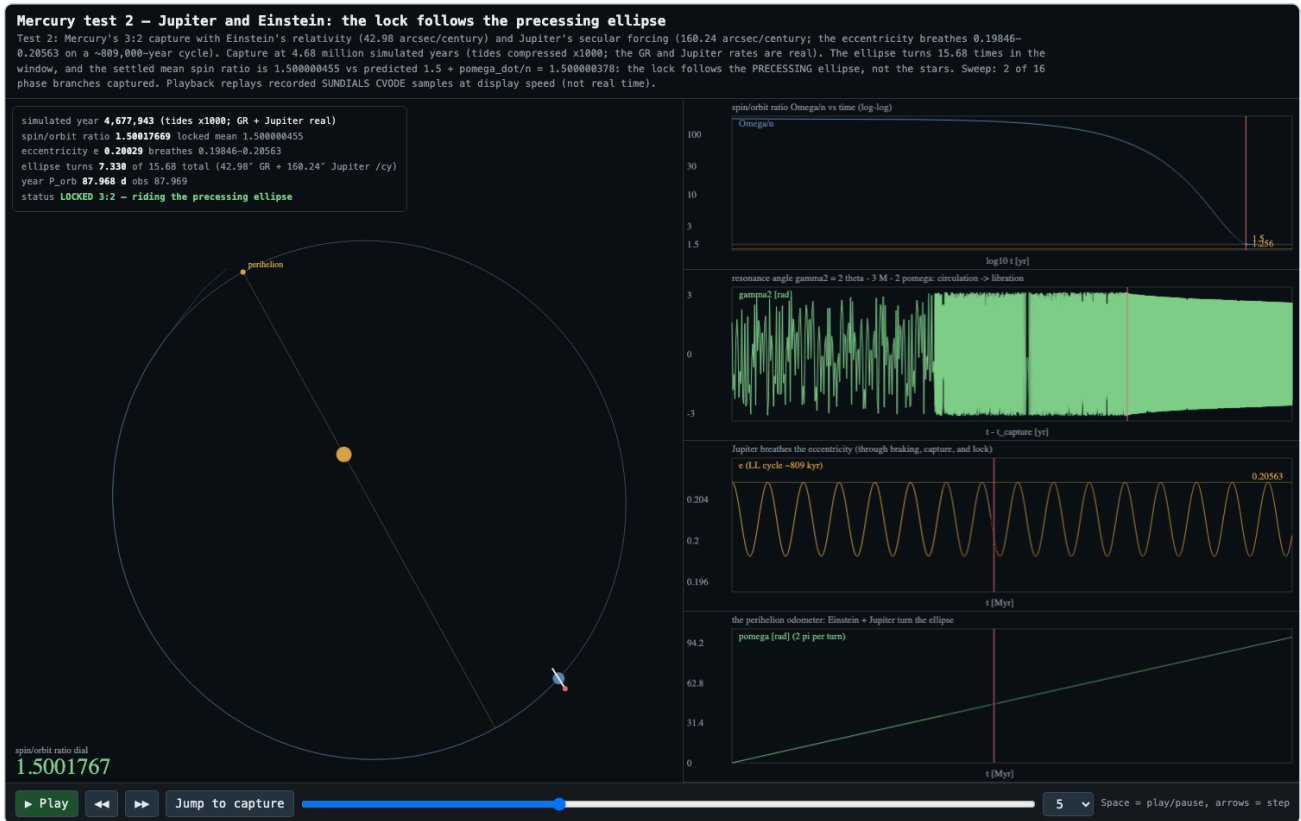
01_planet_mercury_tidal_locking

Planet Mercury's 3:2 spin-orbit capture, computed end to end by the pure-Rust SUNDIALS 7.8.0 CVODE solver: test 1 is the pure two-body story (tidal braking, resonance capture, lock), test 2 adds Einstein's general-relativistic perihelion precession and Jupiter's Laplace-Lagrange secular forcing — the lock follows the precessing ellipse. Each notebook builds its own SQLite database and bakes an animated browser player with a Jump-to-capture button.

mercury_test2_jupiter_gr

This is test 2 of the planet_Mercury project. Test 1 established, with a pure two-body model, that the Sun's tides braked Mercury's spin until the Sun's grip on Mercury's slightly lopsided shape snapped it into the strange 3:2 spin-orbit resonance — three spins for every two trips around the Sun. Test 2 now adds the two pieces of physics test 1 deliberately excluded: 1. Einstein's general-relativistic correction. Near the Sun, gravity is very slightly stronger than Newton's law says. The orbit stays an ellipse, but the ellipse's long axis slowly swings around — the famous 43 arcseconds per century of extra perihelion advance that Newtonian physics could not explain and Einstein's...

- **Executed:** PASS — every cell green (17 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/01_planet_mercury_tidal_locking/mercury_test2_jupiter_gr/mercury_test2_jupiter_gr.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `planet_Mercury/notebook/mercury_test2_jupiter_gr.ipynb`
- **Database:** `planet_Mercury/data/mercury_test2.sqlite3` (SQLite)
- **Player:** python3
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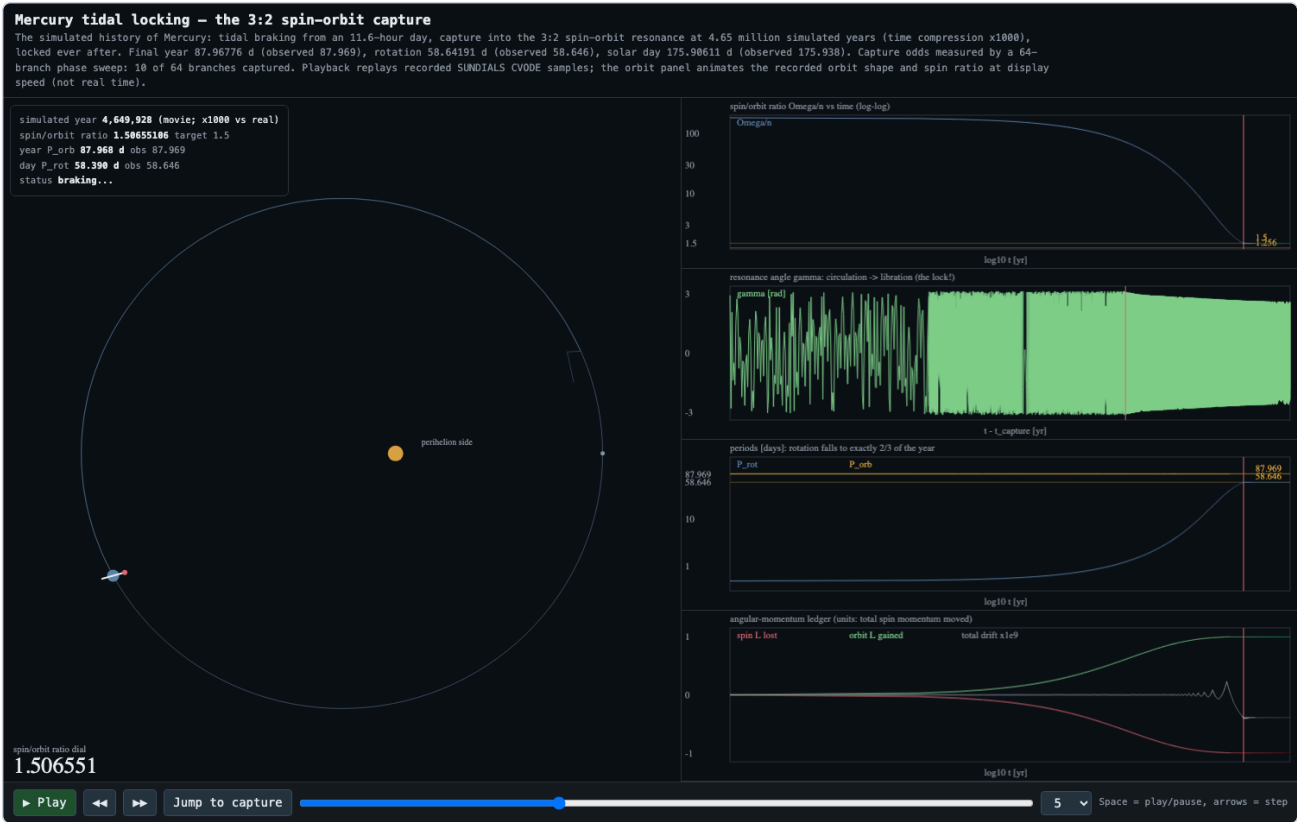


mercury_tidal_locking

Mercury spins on its axis exactly three times for every two trips around the Sun — a "3:2 spin-orbit resonance," which is Mercury's strange form of tidal locking (our Moon, by contrast, is locked 1:1 and always shows Earth one face). This notebook simulates how that happened: the Sun's tides slowly braked Mercury's once-fast spin over billions of years, until the Sun's gravitational grip on Mercury's slightly lopsided shape snapped the spin into step as it fell through the 3:2 ratio.

- **Executed:** PASS — every cell green (18 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/01_planet_mercury_tidal_locking/mercury_tidal_locking/mercury_tidal_locking.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
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- **Database:** `planet_Mercury/data/mercury_orbit.sqlite3` (SQLite)

```
• Player: python3
SolveIt_Notebooks_for_rust/01_planet_mercury_tidal_locking/mercury_tidal_locking/player_mercury_tidal_locking.py
run|gui|jump|capture out.png|data outdir
```



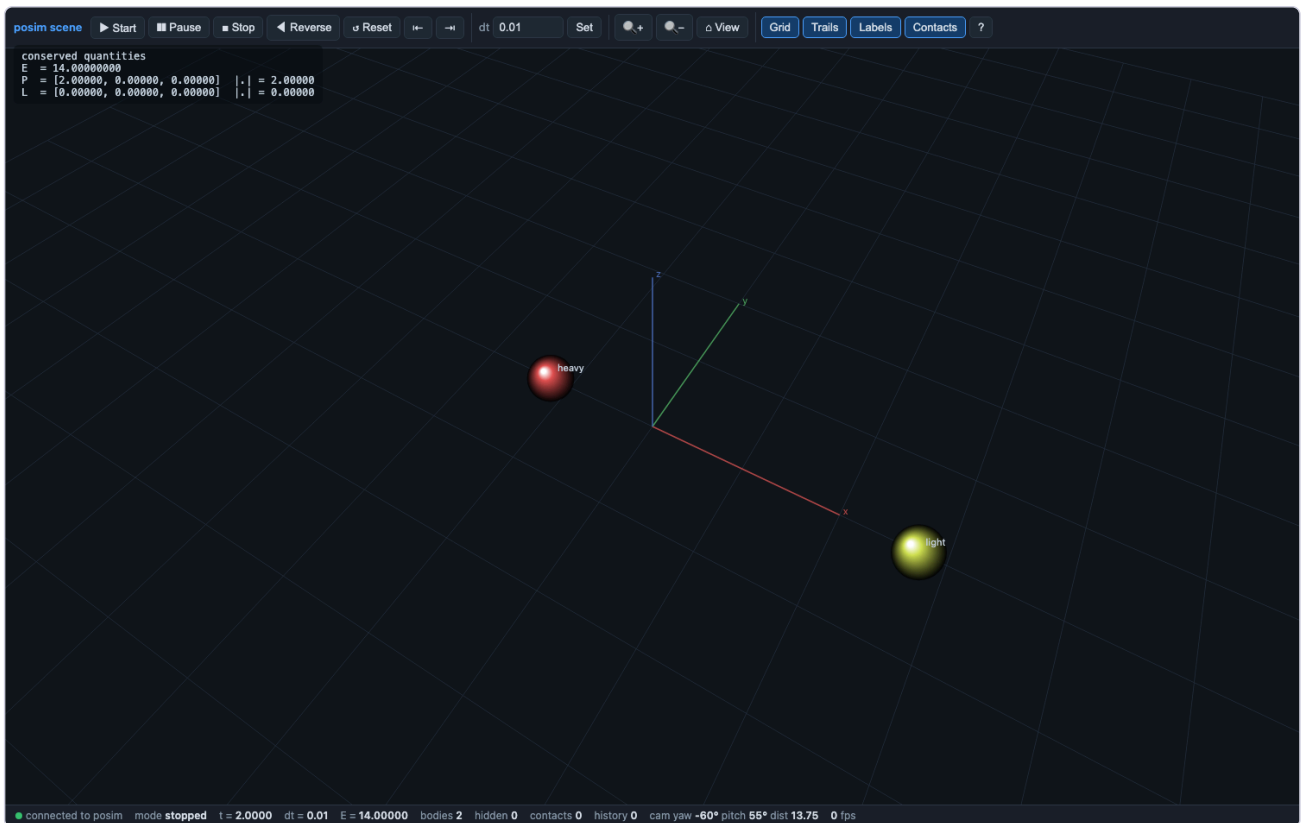
02_solveit_worked_examples

The Solveit worked examples: each notebook drives the posim simulator (pure-Rust SUNDIALS backend) through one classic physics problem with closed-form answers checked in the transcript.

solveit_01_elastic_head_on

This notebook is one half of a pair. The other half is the example file `scripts/solveit/01_elastic_head_on.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

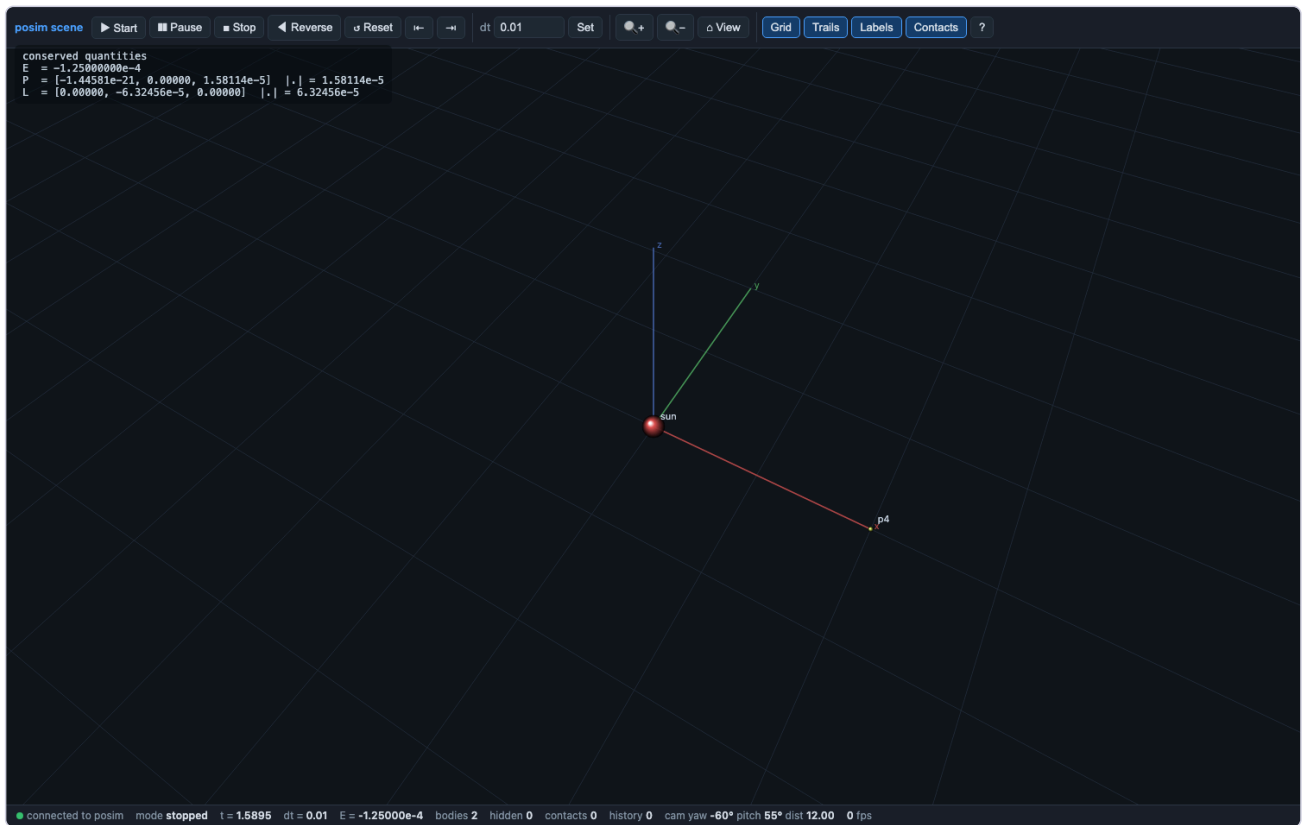
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`run|gui|jump|capture out.png|data outdir`



solveit_02_keplers_third_law

This notebook is one half of a pair. The other half is the example file `scripts/solveit/02_keplers_third_law.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

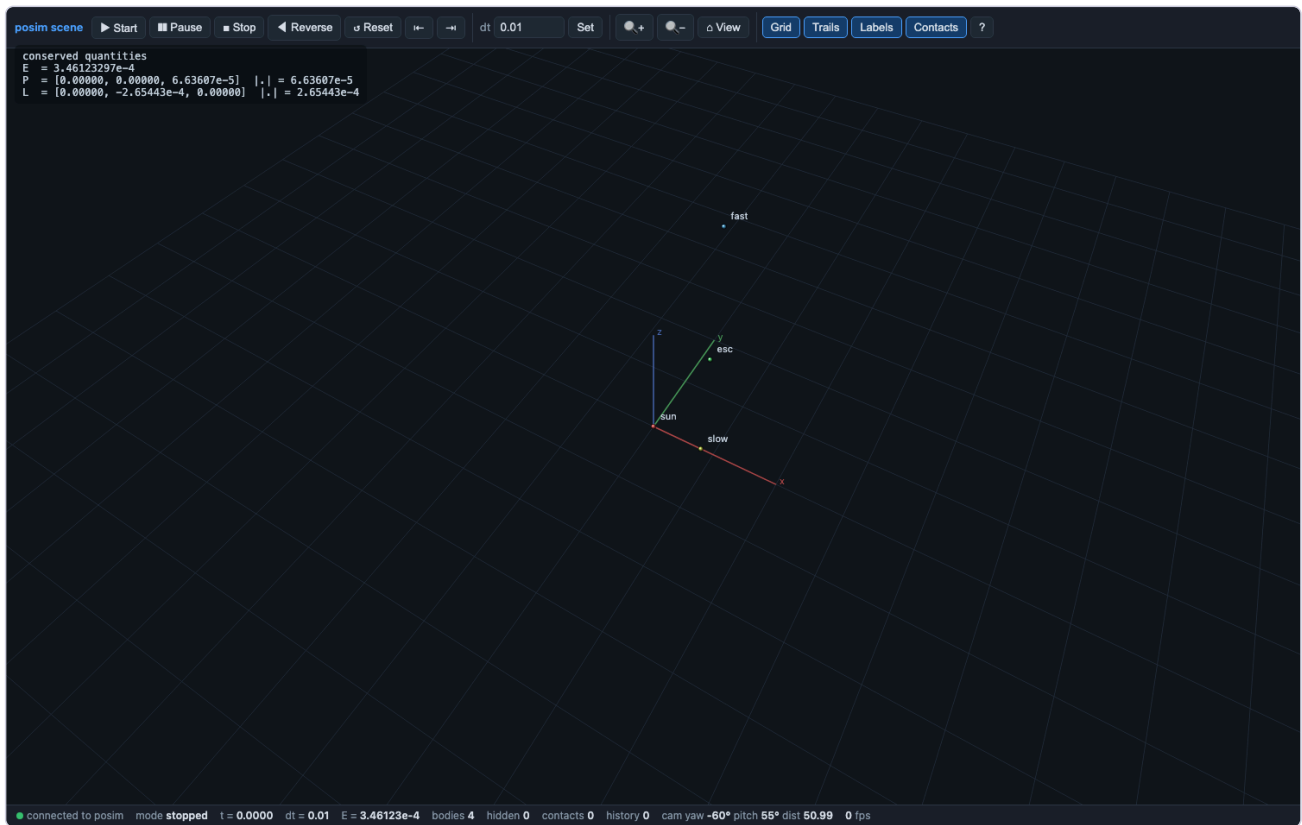
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`run|gui|jump|capture out.png|data outdir`



solveit_03_three_conics

This notebook is one half of a pair. The other half is the example file `scripts/solveit/03_three_conics.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

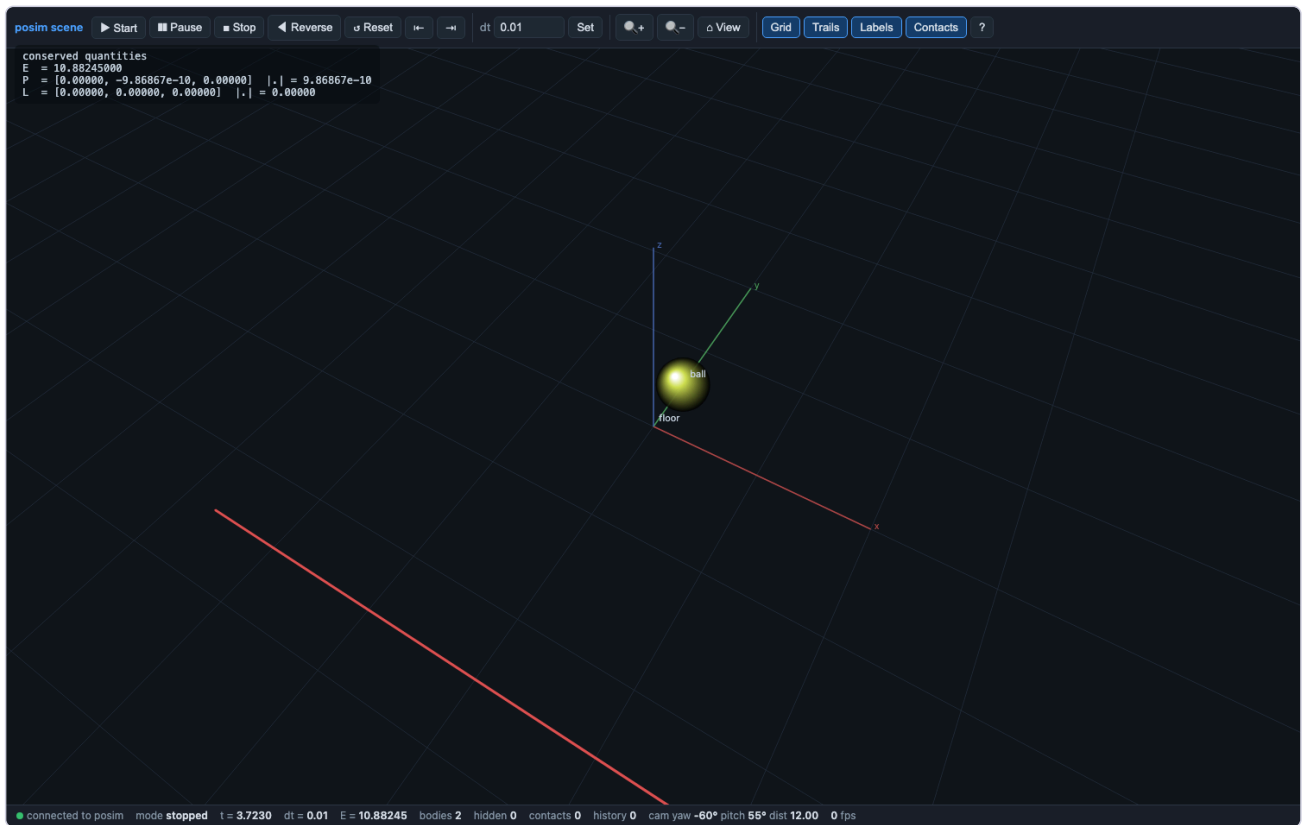
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solveit_04_restitution_ladder

This notebook is one half of a pair. The other half is the example file `scripts/solveit/04_restitution_ladder.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

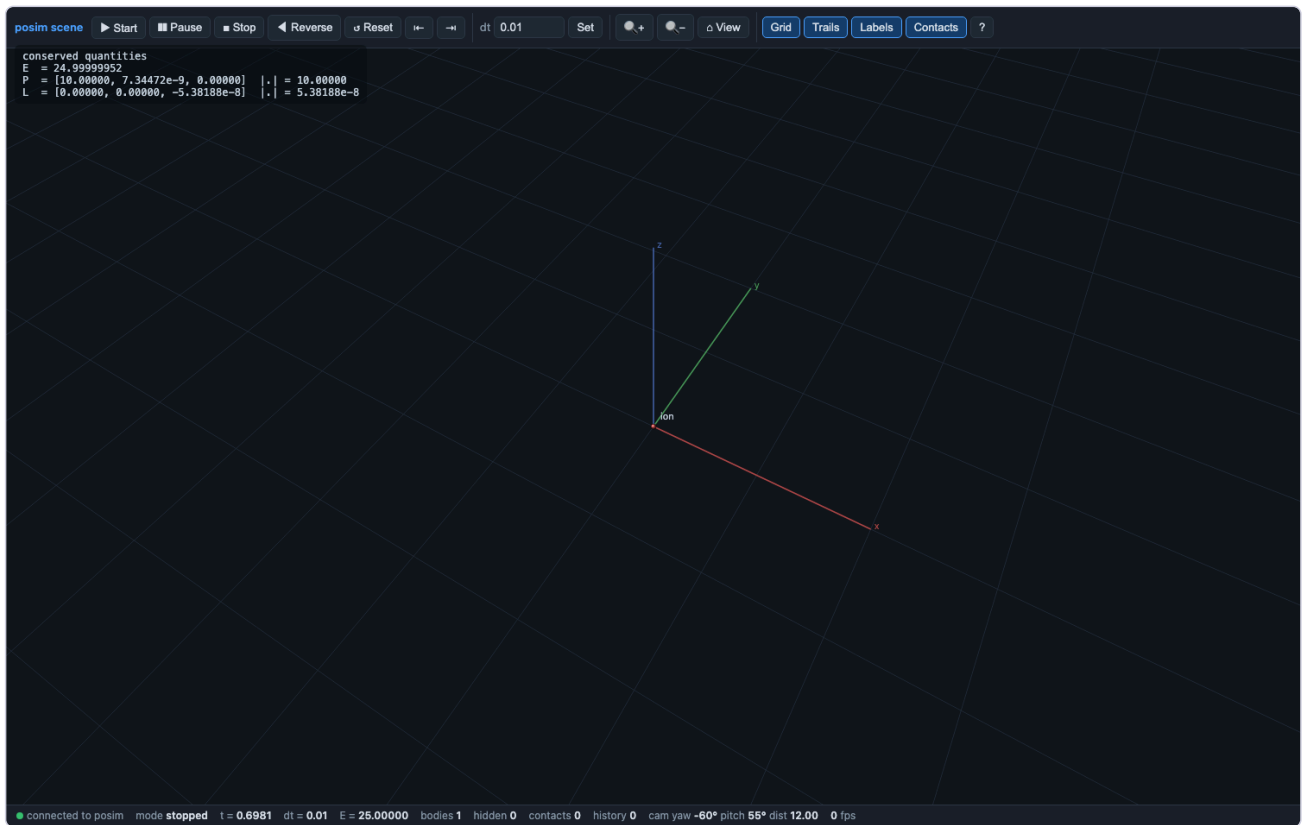
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`run|gui|jump|capture out.png|data outdir`



solveit_05_cyclotron_bdf

This notebook is one half of a pair. The other half is the example file scripts/solveit/05_cyclotron_bdf.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

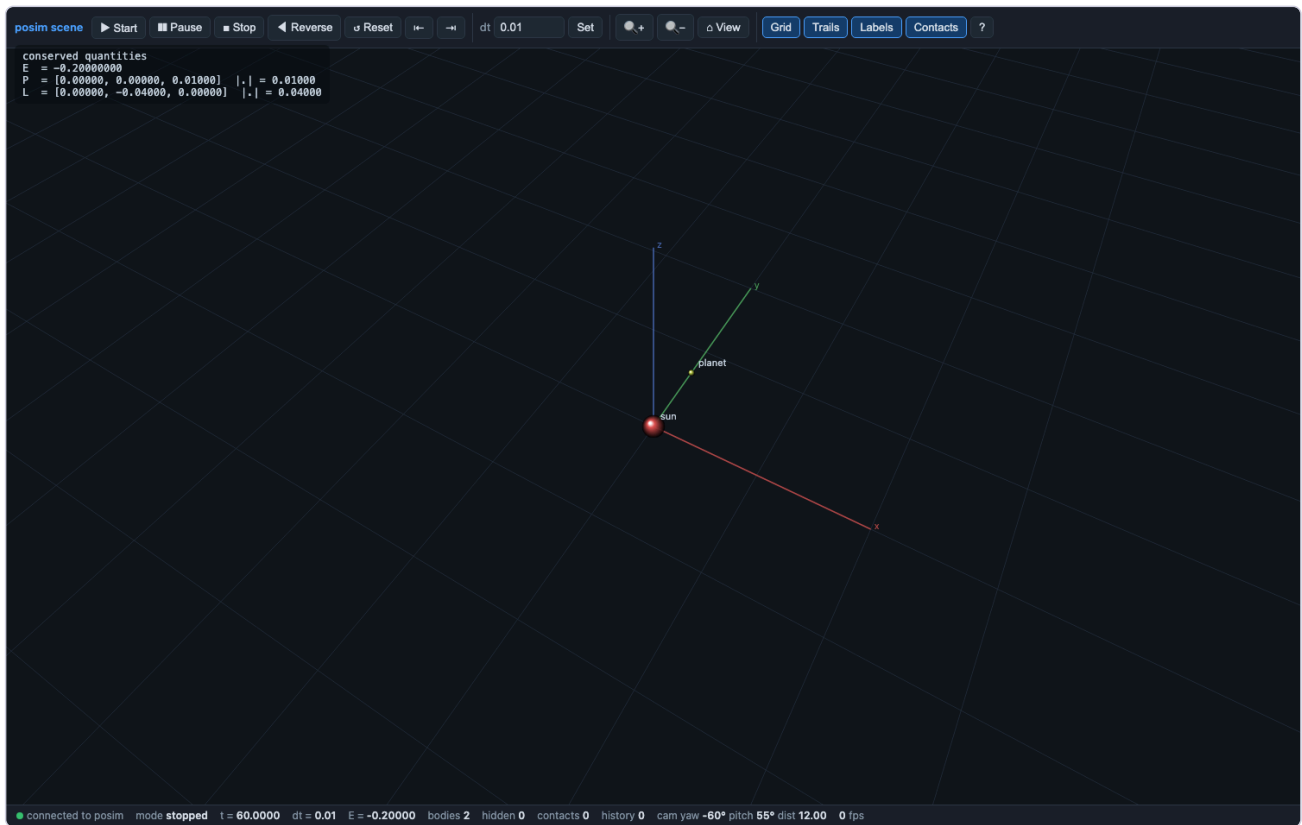
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run|gui|jump|capture out.png|data outdir



solveit_06_symplectic_vs_adaptive

This notebook is one half of a pair. The other half is the example file `scripts/solveit/06_symplectic_vs_adaptive.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

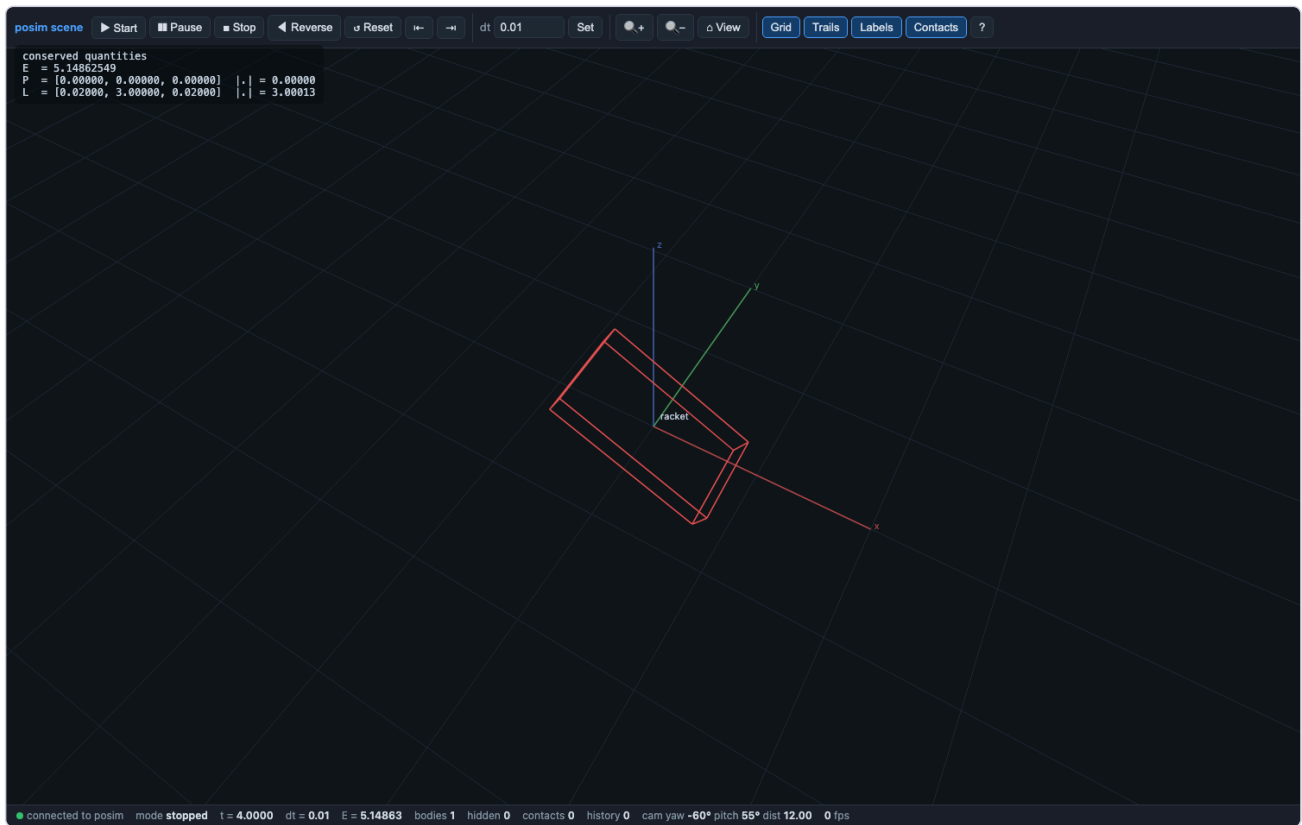
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solveit_07_dzhanibekov

This notebook is one half of a pair. The other half is the example file scripts/solveit/07_dzhanibekov.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

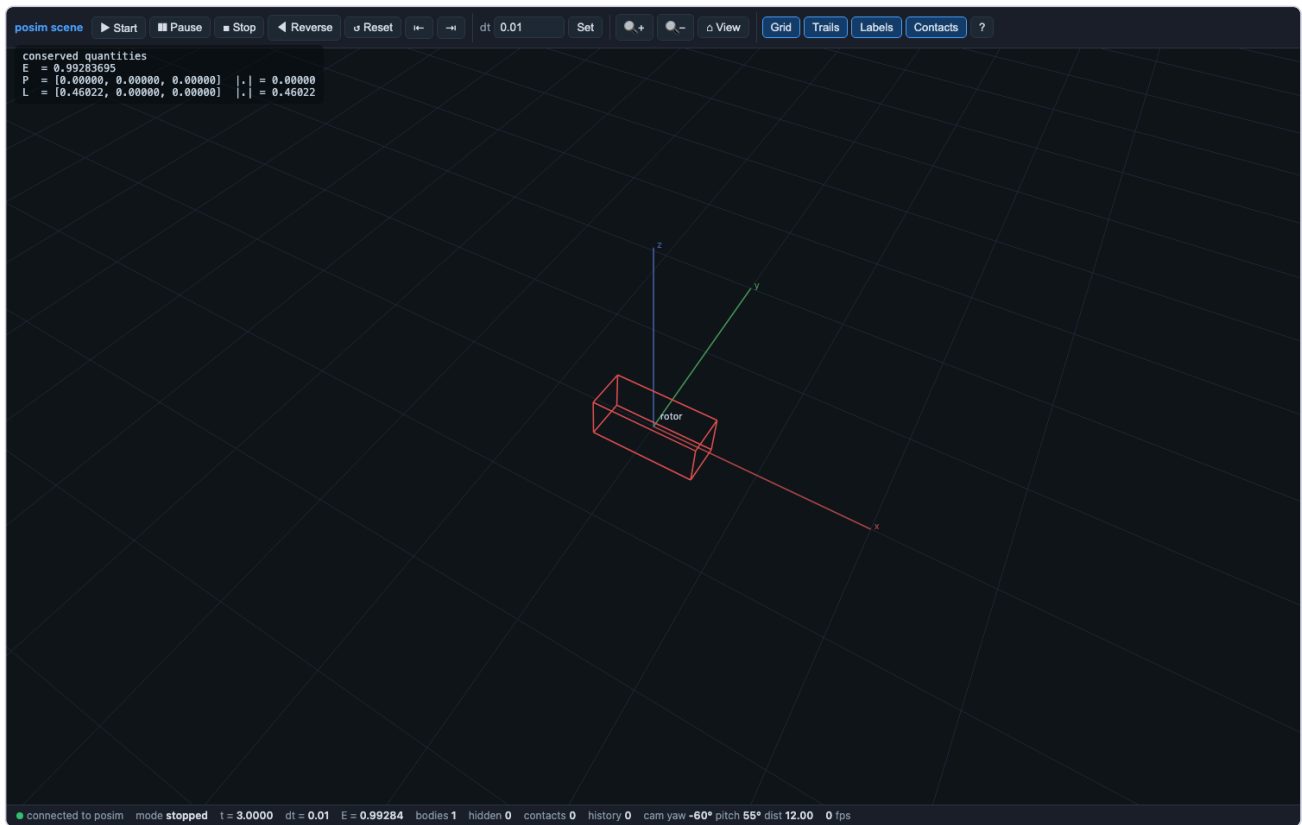
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- **Player:** python3
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run|gui|jump|capture out.png|data outdir



[solveit_08_magnetic_torque](#)

This notebook is one half of a pair. The other half is the example file `scripts/solveit/08_magnetic_torque.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

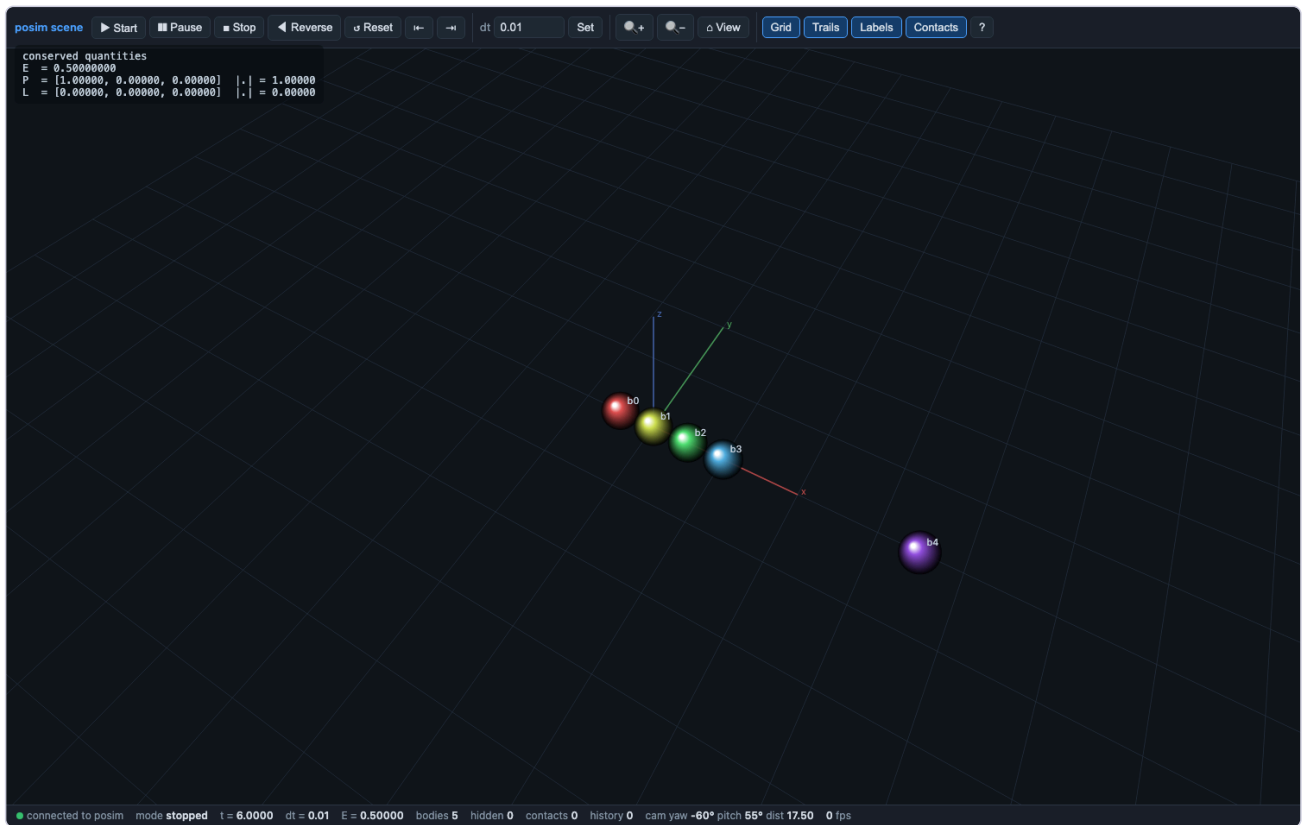
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solveit_09_newtons_cradle

This notebook is one half of a pair. The other half is the example file scripts/solveit/09_newtons_cradle.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

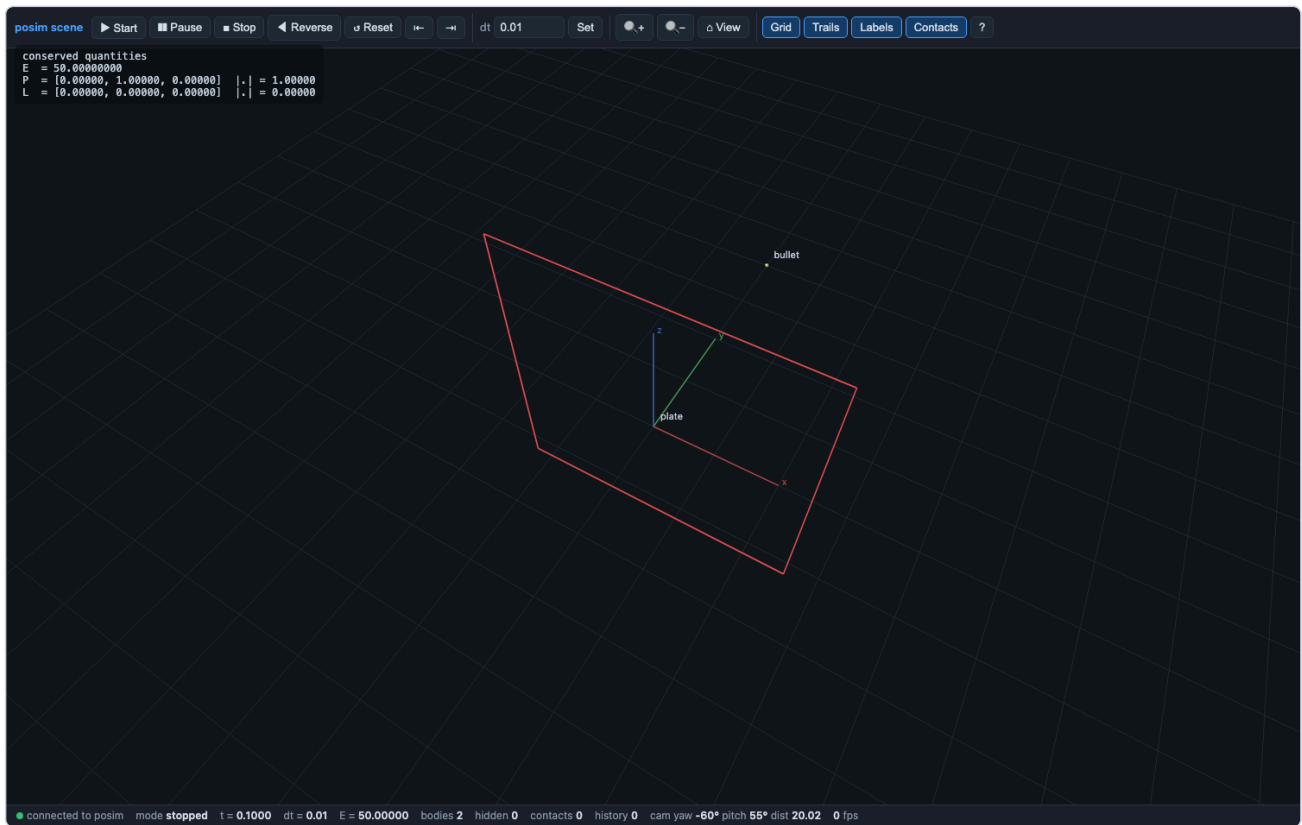
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solveit_10_no_tunnelling

This notebook is one half of a pair. The other half is the example file scripts/solveit/10_no_tunnelling.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

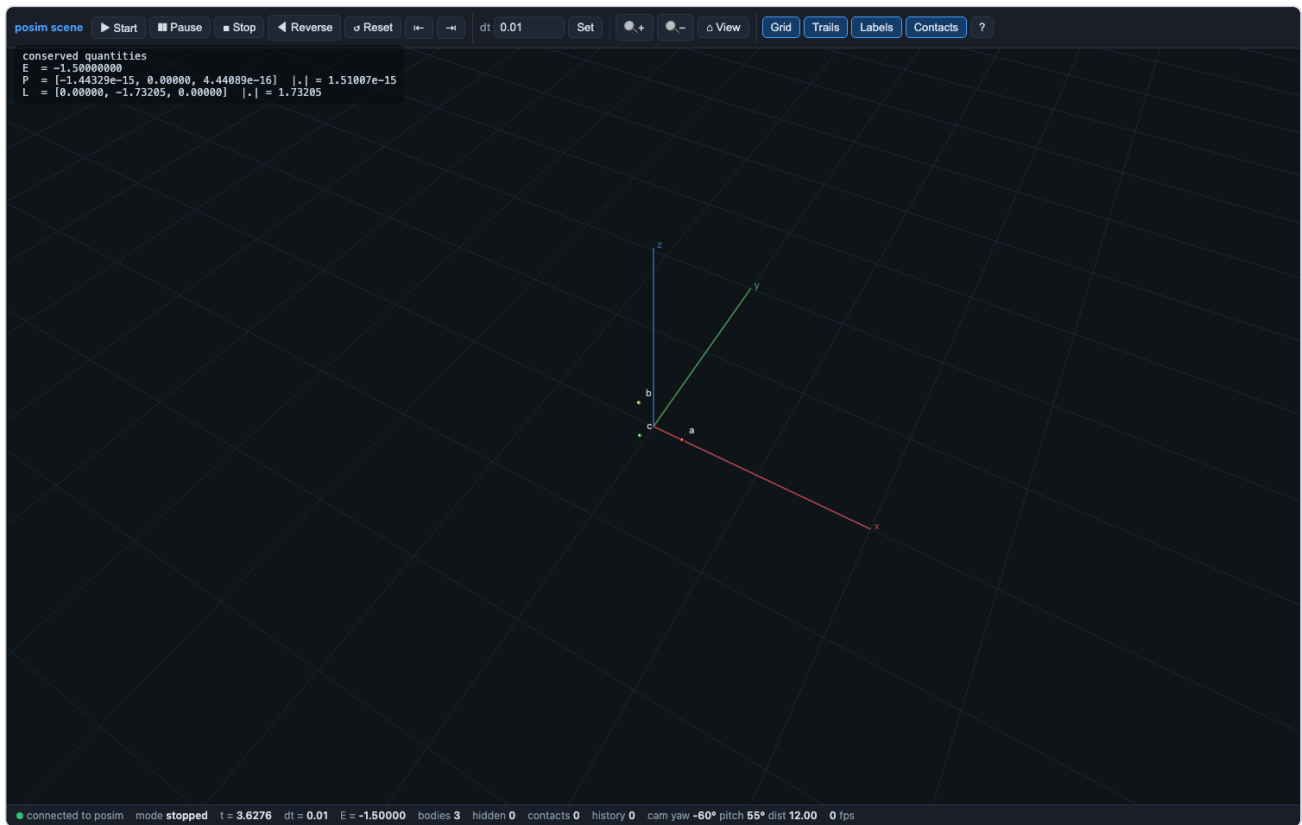
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solveit_11_lagrange_l4

This notebook is one half of a pair. The other half is the example file scripts/solveit/11_lagrange_l4.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

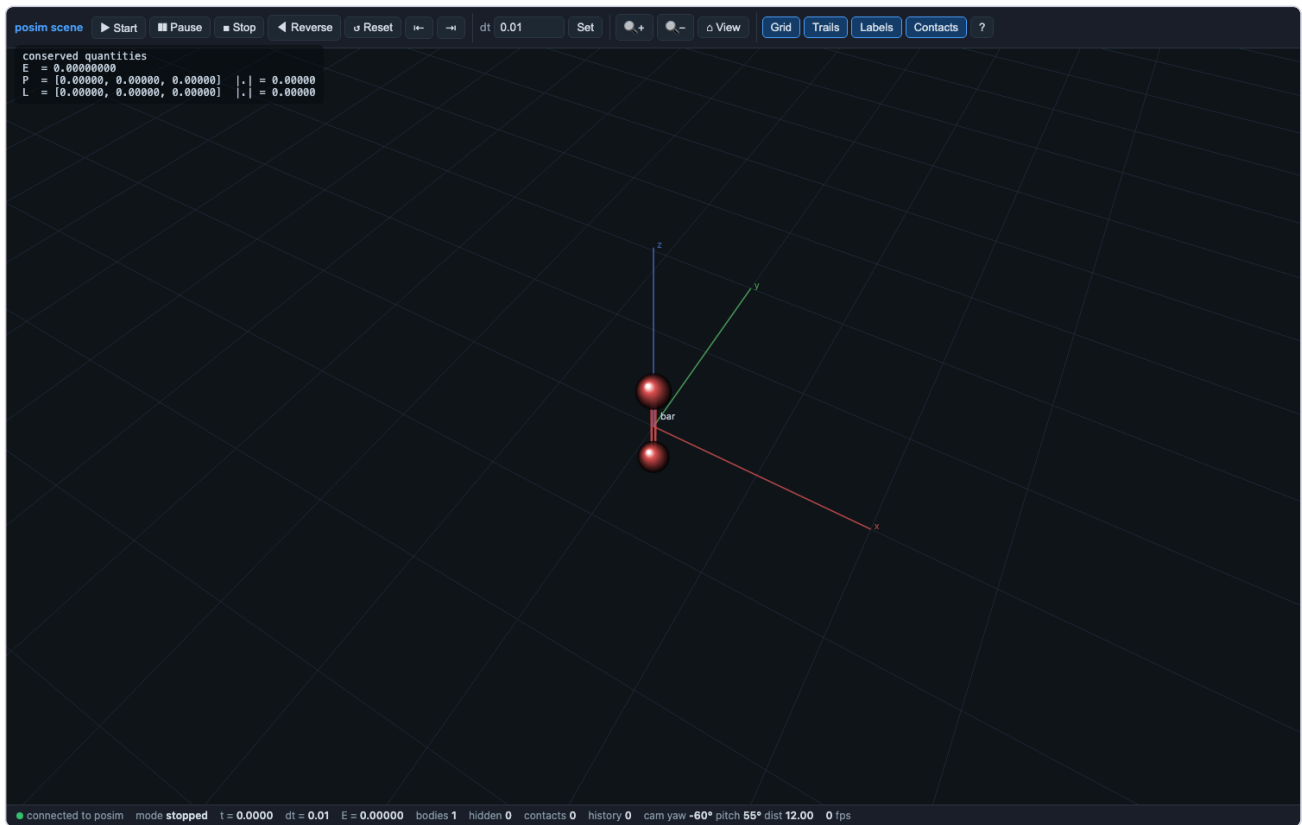
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solveit_12_dumbbell_inertia

This notebook is one half of a pair. The other half is the example file `scripts/solveit/12_dumbbell_inertia.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

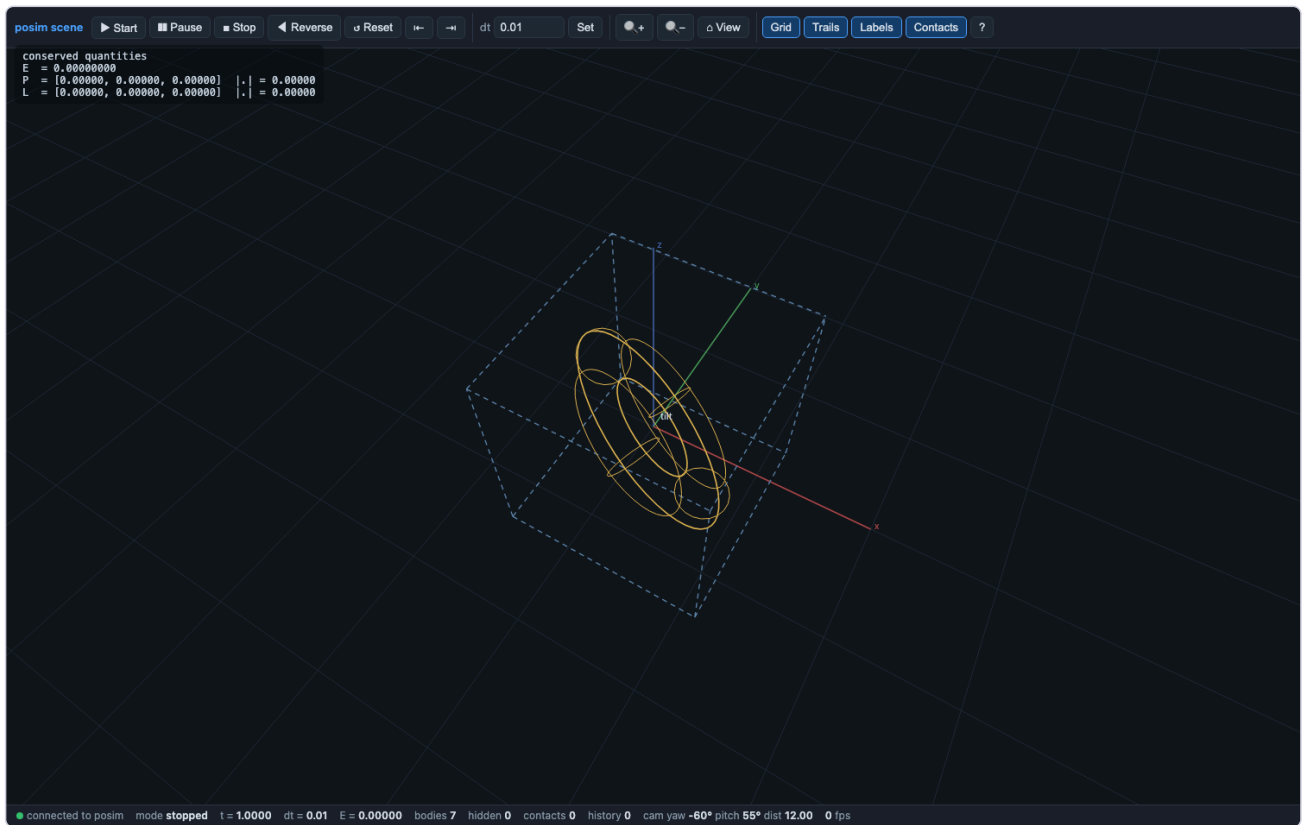
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`run|gui|jump|capture out.png|data outdir`



solveit_13_tilted_torus

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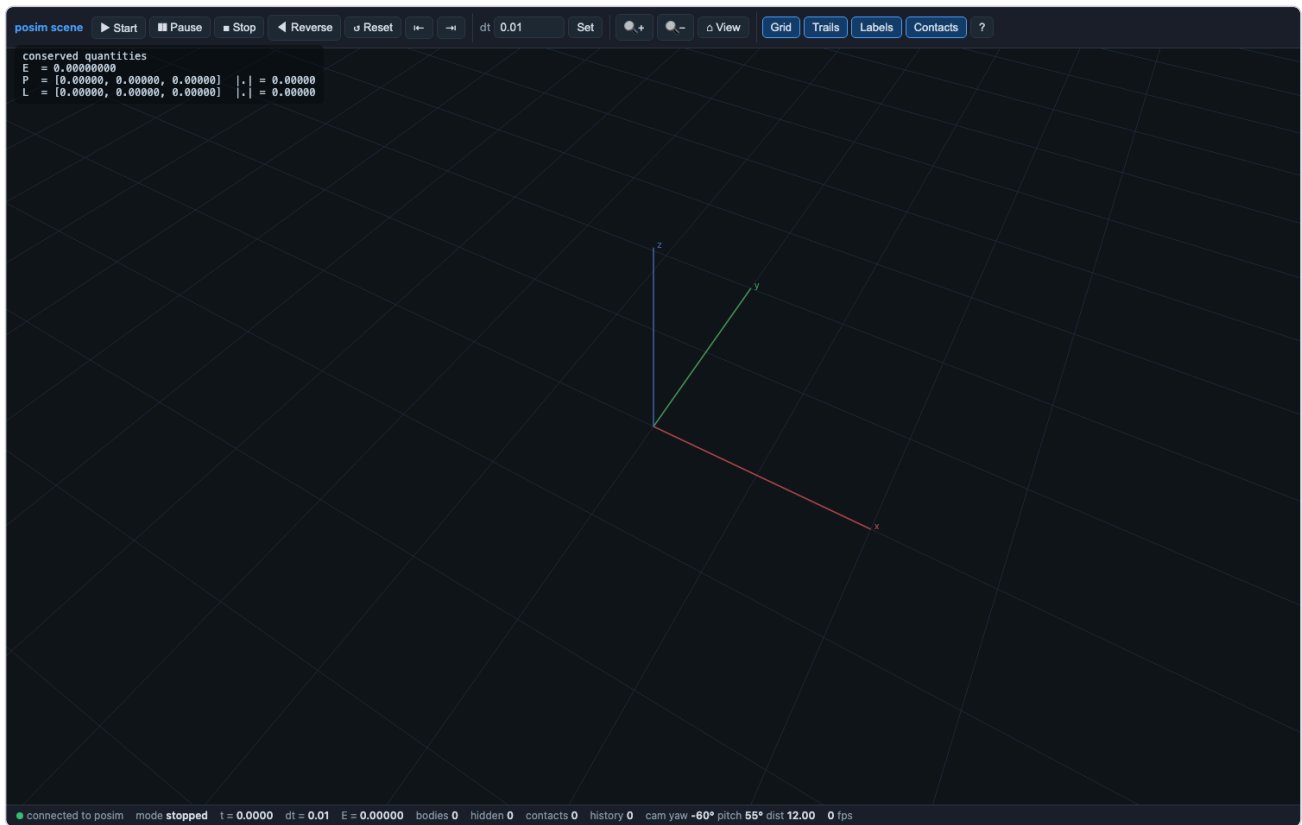
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- **Paired scene/source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/solveit/13_tilted_torus.posim
- **Player:** python3
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run|gui|jump|capture out.png|data outdir



solveit_14_particle_in_a_box

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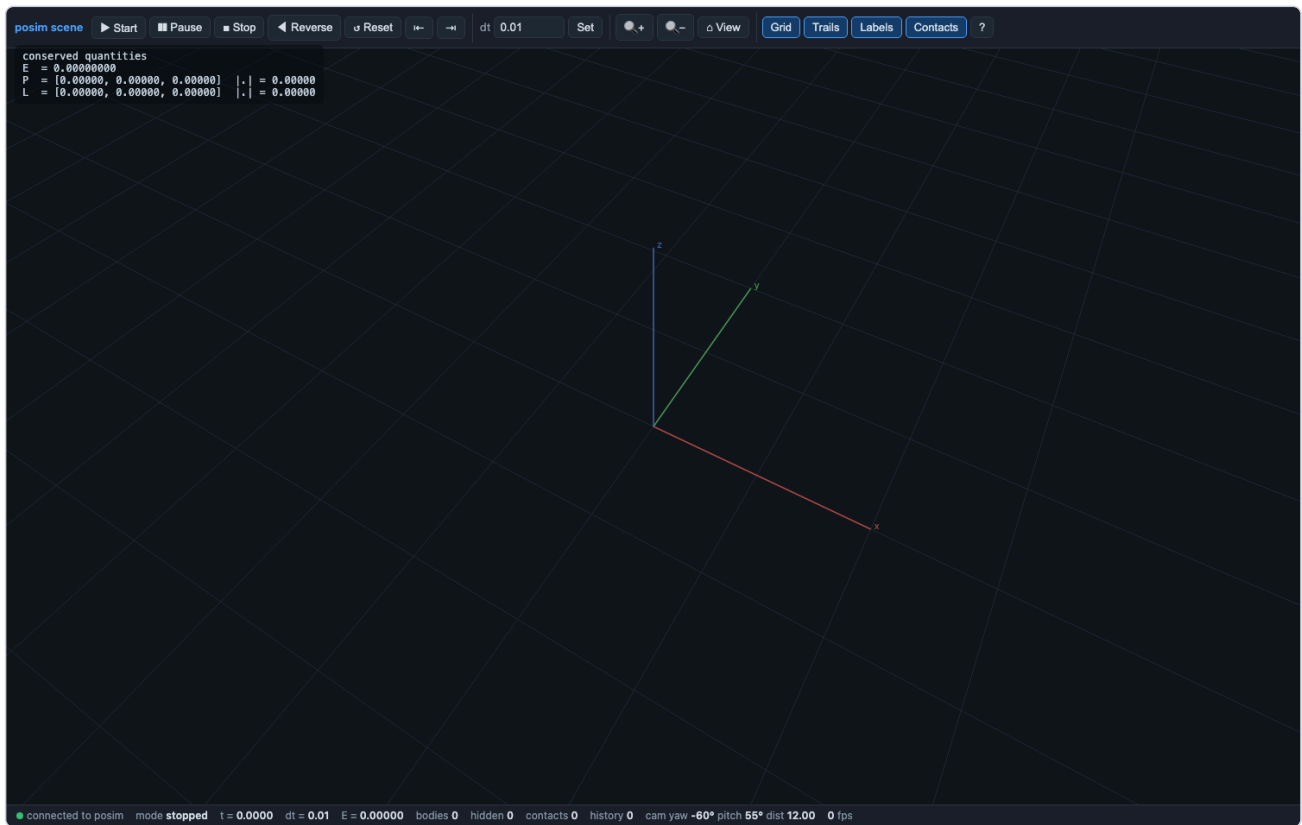
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- **Copy:** `SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_14_particle_in_a_box/solveit_14_particle_in_a_box.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
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- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/solveit/14_particle_in_a_box.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_14_particle_in_a_box/player_solveit_14_particle_in_a_box.py`
`run|gui|jump|capture out.png|data outdir`



solveit_15_tunnelling

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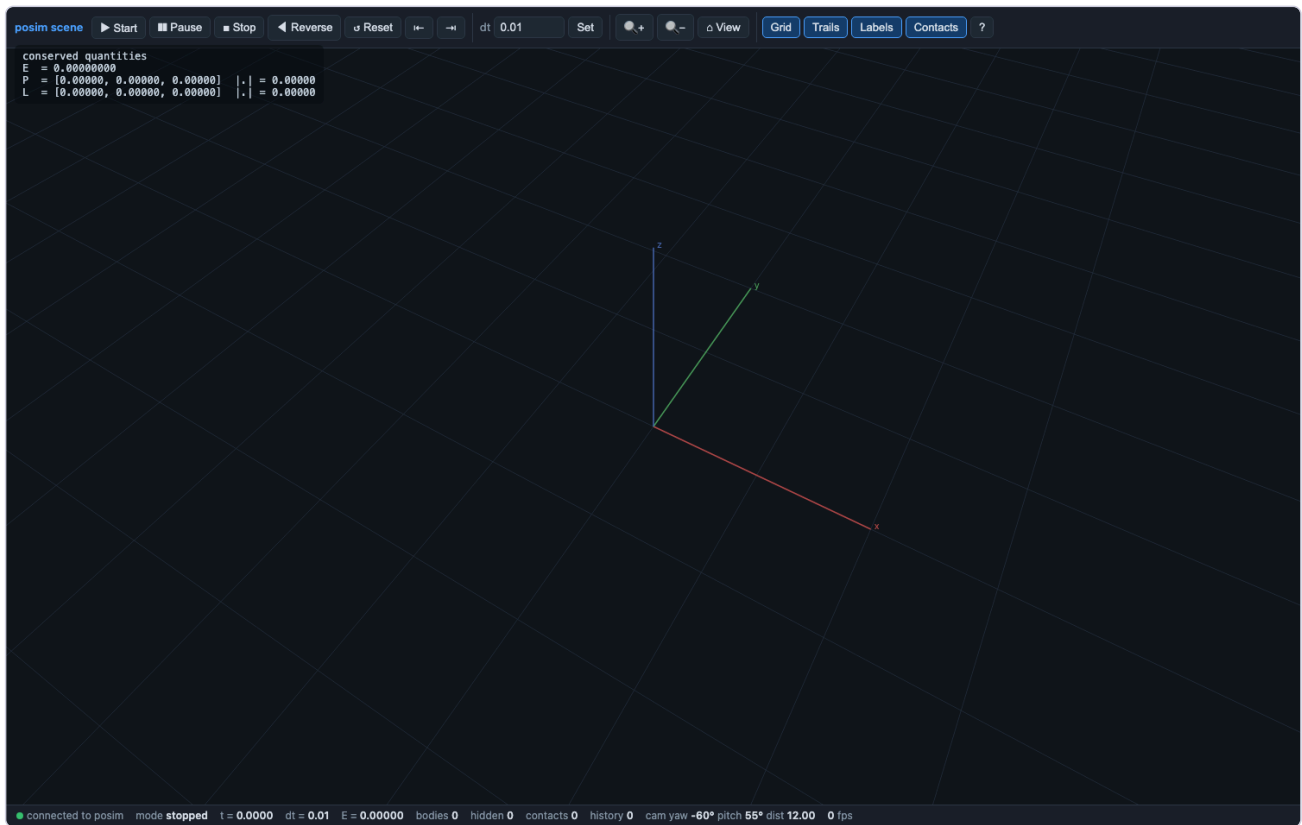
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- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/solveit/15_tunnelling.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_15_tunnelling/player_solveit_15_tunnelling.py run|gui|jump|capture out.png|data outdir`



solveit_16_special_functions

This notebook is one half of a pair. The other half is the example file `scripts/solveit/16_special_functions.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

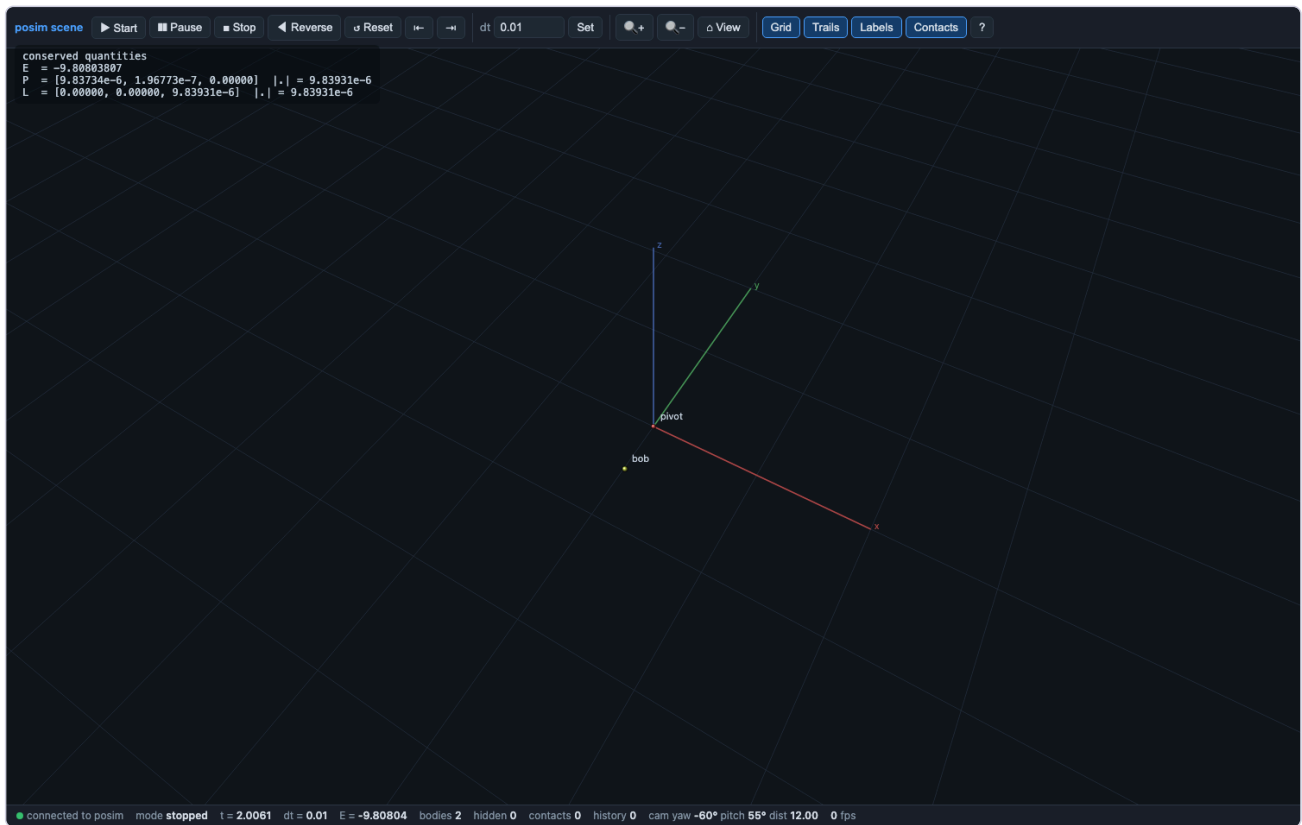
- **Executed:** PASS — every cell green (9 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_16_special_functions/solveit_16_special_functions.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/solveit_16_special_functions.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/solveit/16_special_functions.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_16_special_functions/player_solveit_16_special_functions.py`
`run|gui|jump|capture out.png|data outdir`



solveit_17_pendulum_dae

This notebook is one half of a pair. The other half is the example file scripts/solveit/17_pendulum_dae.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

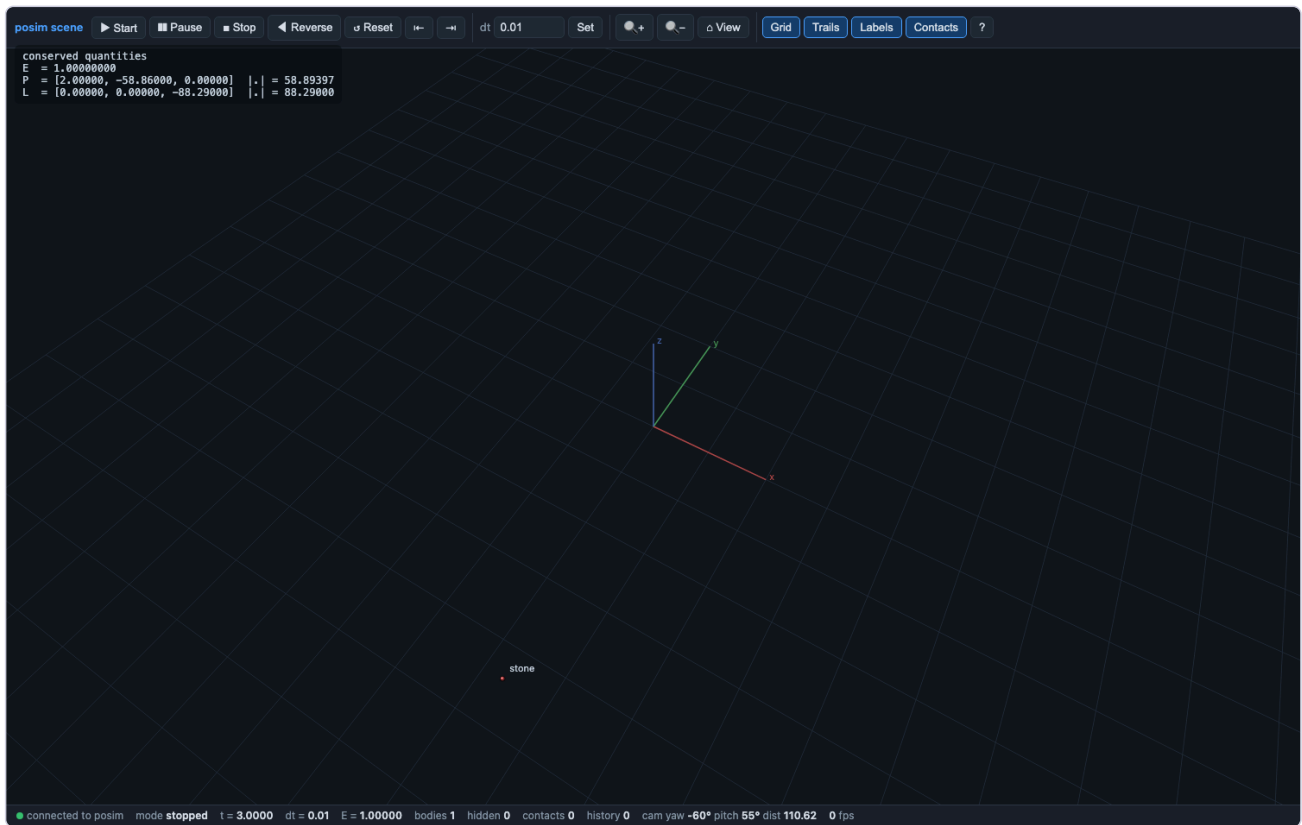
- **Executed:** PASS — every cell green (10 code cells)
- **Copy:** SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_17_pendulum_dae/solveit_17_pendulum_dae.ipynb (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/solveit_17_pendulum_dae.ipynb
- **Paired scene/source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/solveit/17_pendulum_dae.posim
- **Player:** python3
SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_17_pendulum_dae/player_solveit_17_pendulum_dae.py
run|gui|jump|capture out.png|data outdir



solveit_18_equilibrium_and_sensitivity

This notebook is one half of a pair. The other half is the example file `scripts/solveit/18_equilibrium_and_sensitivity.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

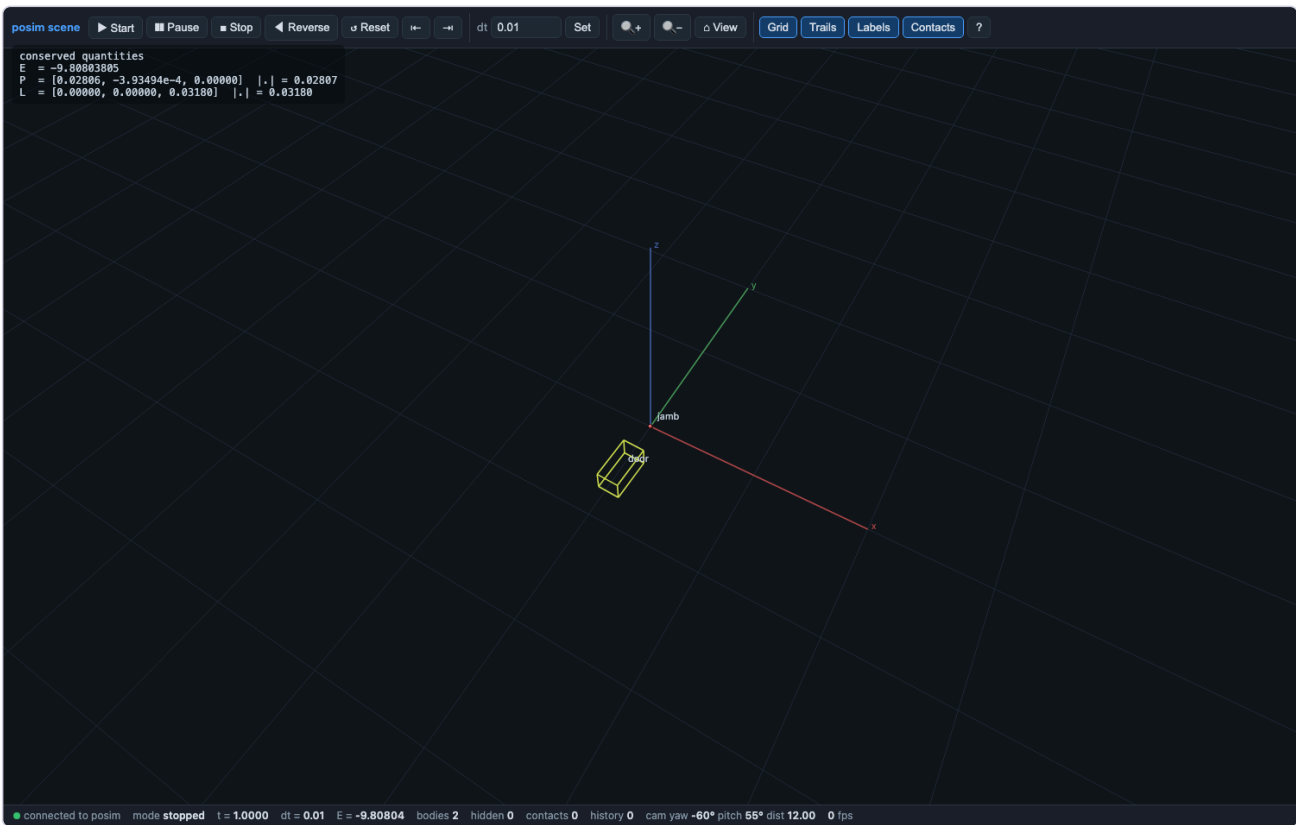
- **Executed:** PASS — every cell green (12 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_18_equilibrium_and_sensitivity/solveit_18_equilibrium_and_sensitivity.i`
 (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/solveit_18_equilibrium_and_sensitivity.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/solveit/18_equilibrium_and_sensitivity.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_18_equilibrium_and_sensitivity/player_solveit_18_equilibrium_and_sensitivity.py`
`run|gui|jump|capture out.png|data outdir`



solveit_19_hinged_door

This notebook is one half of a pair. The other half is the example file scripts/solveit/19_hinged_door.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (10 code cells)
- **Copy:** SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_19_hinged_door/solveit_19_hinged_door.ipynb (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/solveit_19_hinged_door.ipynb
- **Paired scene/source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/solveit/19_hinged_door.posim
- **Player:** python3
SolveIt_Notebooks_for_rust/02_solveit_worked_examples/solveit_19_hinged_door/player_solveit_19_hinged_door.py
run|gui|jump|capture out.png|data outdir



03_dynamics_and_routh

The dynamic notebooks — free rigid-body motion, charged particles in fields, orbital mechanics, and the 34 Routh rigid-body problems — each a machine-mode posim session with its analytic checks.

dynamic_billiard_box

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/billiard_box.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (7 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_billiard_box/dynamic_billiard_box.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_billiard_box.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/billiard_box.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_billiard_box/player_dynamic_billiard_box.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_billiard_break

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/billiard_break.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (6 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_billiard_break/dynamic_billiard_break.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_billiard_break.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/billiard_break.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_billiard_break/player_dynamic_billiard_break.py`
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dynamic_bouncing_ball_restitution

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/bouncing_ball_restitution.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (10 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_bouncing_ball_restitution/dynamic_bouncing_ball_restitution.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_bouncing_ball_restitution.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/bouncing_ball_restitution.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_bouncing_ball_restitution/player_dynamic_bouncing_ball_restitution.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_box_of_shapes

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/box_of_shapes.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (11 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_box_of_shapes/dynamic_box_of_shapes.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_box_of_shapes.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/box_of_shapes.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_box_of_shapes/player_dynamic_box_of_shapes.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_box_of_shapes_m32

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/box_of_shapes_m32.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (11 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_box_of_shapes_m32/dynamic_box_of_shapes_m32.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_box_of_shapes_m32.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/box_of_shapes_m32.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_box_of_shapes_m32/player_dynamic_box_of_shapes_m32.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_charged_in_b_field

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/charged_in_b_field.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (10 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_charged_in_b_field/dynamic_charged_in_b_field.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_charged_in_b_field.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/charged_in_b_field.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_charged_in_b_field/player_dynamic_charged_in_b_field.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_charged_in_e_field

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/charged_in_e_field.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (8 code cells)
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- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_charged_in_e_field.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/charged_in_e_field.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_charged_in_e_field/player_dynamic_charged_in_e_field.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_colliding_binary

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/colliding_binary.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (6 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_colliding_binary/dynamic_colliding_binary.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_colliding_binary.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/colliding_binary.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_colliding_binary/player_dynamic_colliding_binary.py`
`run|gui|jump|capture out.png|data outdir`



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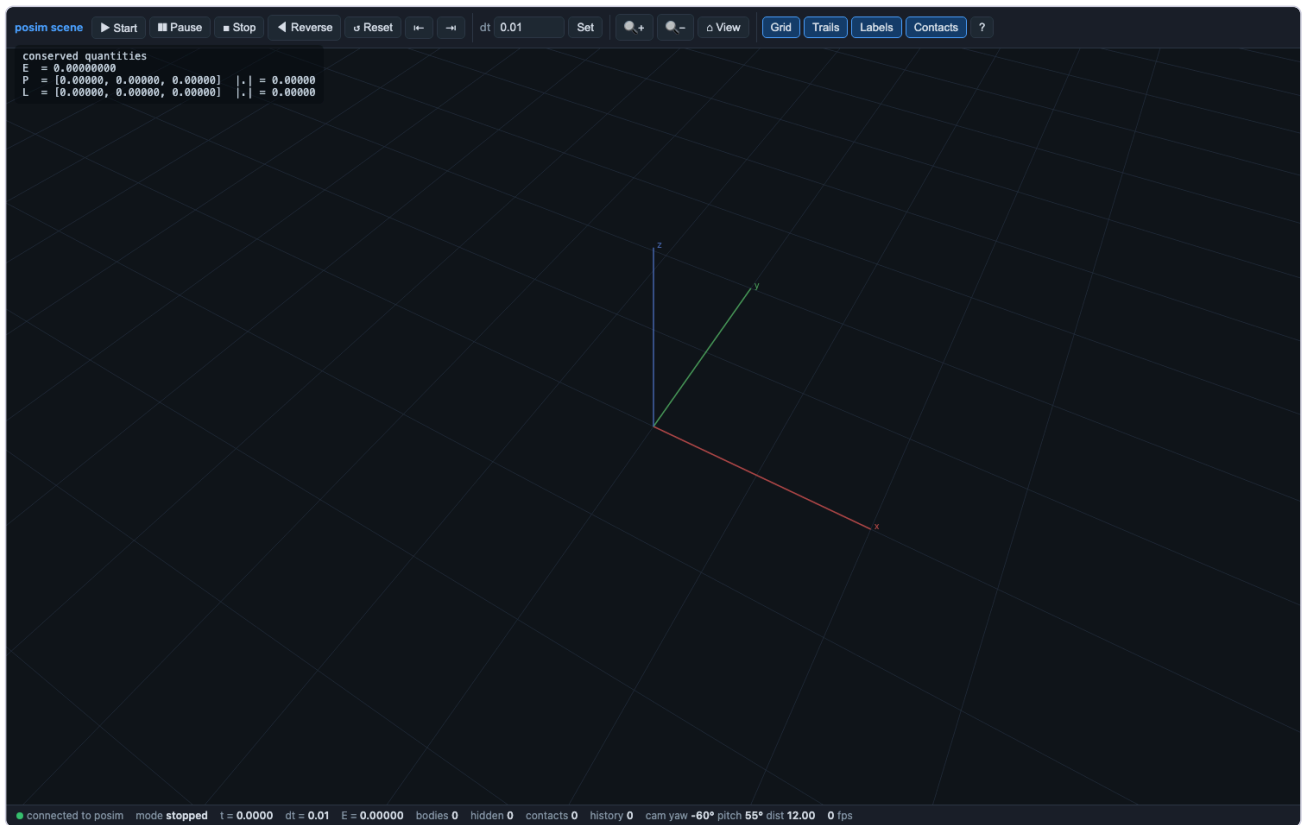
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dynamic_double_slit

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/double_slit.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (17 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_double_slit/dynamic_double_slit.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_double_slit.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/double_slit.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_double_slit/player_dynamic_double_slit.py`
`run|gui|jump|capture out.png|data outdir`



dynamic_head_on_exchange

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/head_on_exchange.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (7 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_head_on_exchange/dynamic_head_on_exchange.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_head_on_exchange.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/head_on_exchange.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_head_on_exchange/player_dynamic_head_on_exchange.py run|gui|jump|capture out.png|data outdir`



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dynamic_kepler_orbit

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/kepler_orbit.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (6 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_kepler_orbit/dynamic_kepler_orbit.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_kepler_orbit.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/kepler_orbit.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_kepler_orbit/player_dynamic_kepler_orbit.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_magnetic_spin_up

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/magnetic_spin_up.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (7 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_magnetic_spin_up/dynamic_magnetic_spin_up.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_magnetic_spin_up.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/magnetic_spin_up.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_magnetic_spin_up/player_dynamic_magnetic_spin_up.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_newtons_cradle

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/newtons_cradle.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (7 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_newtons_cradle/dynamic_newtons_cradle.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_newtons_cradle.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/newtons_cradle.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_newtons_cradle/player_dynamic_newtons_cradle.py run|gui|jump|capture out.png|data outdir`



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dynamic_outer_solar_system

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/outer_solar_system.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (7 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_outer_solar_system/dynamic_outer_solar_system.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_outer_solar_system.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/outer_solar_system.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_outer_solar_system/player_dynamic_outer_solar_system.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_restitution_ladder

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/restitution_ladder.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (10 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_restitution_ladder/dynamic_restitution_ladder.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_restitution_ladder.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/restitution_ladder.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_restitution_ladder/player_dynamic_restitution_ladder.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_double_star_period

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_double_star_period.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (27 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_double_star_period/dynamic_routh_double_star_period.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_double_star_period.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_double_star_period.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_double_star_period/player_dynamic_routh_double_star_period.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_apsidal_symmetry

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_apsidal_symmetry.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (19 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_apsidal_symmetry/dynamic_routh_p1_apsidal_symmetry.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_apsidal_symmetry.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_apsidal_symmetry.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_apsidal_symmetry/player_dynamic_routh_p1_apsidal_symmetry.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_centre_of_gravity

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_centre_of_gravity.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (12 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_centre_of_gravity/dynamic_routh_p1_centre_of_gravity.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_centre_of_gravity.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_centre_of_gravity.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_centre_of_gravity/player_dynamic_routh_p1_centre_of_gravity.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_collinear_three_body

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_collinear_three_body.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (18 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_collinear_three_body/dynamic_routh_p1_collinear_three_body.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_collinear_three_body.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_collinear_three_body.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_collinear_three_body/player_dynamic_routh_p1_collinear_three_body.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_equal_periods

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_equal_periods.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (15 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_equal_periods/dynamic_routh_p1_equal_periods.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_equal_periods.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_equal_periods.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_equal_periods/player_dynamic_routh_p1_equal_periods.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_equilateral_three_body

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_equilateral_three_body.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (19 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_equilateral_three_body/dynamic_routh_p1_equilateral_three_body.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_equilateral_three_body.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_equilateral_three_body.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_equilateral_three_body/player_dynamic_routh_p1_equilateral_three_boc`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_escape_velocity

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_escape_velocity.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (12 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_escape_velocity/dynamic_routh_p1_escape_velocity.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_escape_velocity.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_escape_velocity.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_escape_velocity/player_dynamic_routh_p1_escape_velocity.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_expanding_sphere

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_expanding_sphere.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (13 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_expanding_sphere/dynamic_routh_p1_expanding_sphere.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_expanding_sphere.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_expanding_sphere.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_expanding_sphere/player_dynamic_routh_p1_expanding_sphere.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_geometric_progression

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_geometric_progression.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (12 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_geometric_progression/dynamic_routh_p1_geometric_progression.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_geometric_progression.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_geometric_progression.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_geometric_progression/player_dynamic_routh_p1_geometric_progression.run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_hodograph_circle

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_hodograph_circle.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (15 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_hodograph_circle/dynamic_routh_p1_hodograph_circle.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_hodograph_circle.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_hodograph_circle.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_hodograph_circle/player_dynamic_routh_p1_hodograph_circle.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_kepler_equation

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_kepler_equation.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (15 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_kepler_equation/dynamic_routh_p1_kepler_equation.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_kepler_equation.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_kepler_equation.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_kepler_equation/player_dynamic_routh_p1_kepler_equation.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_lambert_theorem

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_lambert_theorem.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (15 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_lambert_theorem/dynamic_routh_p1_lambert_theorem.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_lambert_theorem.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_lambert_theorem.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_lambert_theorem/player_dynamic_routh_p1_lambert_theorem.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_oblique_impact

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_oblique_impact.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (13 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_oblique_impact/dynamic_routh_p1_oblique_impact.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_oblique_impact.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_oblique_impact.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_oblique_impact/player_dynamic_routh_p1_oblique_impact.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_parabola_of_safety

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_parabola_of_safety.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (21 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_parabola_of_safety/dynamic_routh_p1_parabola_of_safety.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_parabola_of_safety.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_parabola_of_safety.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_parabola_of_safety/player_dynamic_routh_p1_parabola_of_safety.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_sphere_exchange

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_sphere_exchange.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (14 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_sphere_exchange/dynamic_routh_p1_sphere_exchange.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_sphere_exchange.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_sphere_exchange.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_sphere_exchange/player_dynamic_routh_p1_sphere_exchange.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_three_projectiles

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_three_projectiles.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (14 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_three_projectiles/dynamic_routh_p1_three_projectiles.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_three_projectiles.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_three_projectiles.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_three_projectiles/player_dynamic_routh_p1_three_projectiles.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p1_two_trajectories

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p1_two_trajectories.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (17 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_two_trajectories/dynamic_routh_p1_two_trajectories.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p1_two_trajectories.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p1_two_trajectories.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p1_two_trajectories/player_dynamic_routh_p1_two_trajectories.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_correlated_bodies

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_correlated_bodies.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (15 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_correlated_bodies/dynamic_routh_p2_correlated_bodies.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_correlated_bodies.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_correlated_bodies.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_correlated_bodies/player_dynamic_routh_p2_correlated_bodies.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_fixed_couple

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_fixed_couple.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (12 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_fixed_couple/dynamic_routh_p2_fixed_couple.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_fixed_couple.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_fixed_couple.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_fixed_couple/player_dynamic_routh_p2_fixed_couple.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_impulsive_couple

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_impulsive_couple.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (14 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_impulsive_couple/dynamic_routh_p2_impulsive_couple.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_impulsive_couple.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_impulsive_couple.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_impulsive_couple/player_dynamic_routh_p2_impulsive_couple.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_invariable_line

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_invariable_line.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (14 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_invariable_line/dynamic_routh_p2_invariable_line.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_invariable_line.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_invariable_line.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_invariable_line/player_dynamic_routh_p2_invariable_line.py`
`run|gui|jump|capture out.png|data outdir`



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Details

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dynamic_routh_p2_mean_axis_instability

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_mean_axis_instability.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (19 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_mean_axis_instability/dynamic_routh_p2_mean_axis_instability.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_mean_axis_instability.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_mean_axis_instability.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_mean_axis_instability/player_dynamic_routh_p2_mean_axis_instability.run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_period_ratio

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_period_ratio.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (13 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_period_ratio/dynamic_routh_p2_period_ratio.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_period_ratio.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_period_ratio.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_period_ratio/player_dynamic_routh_p2_period_ratio.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_poinsot_rolling

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_poinsot_rolling.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (15 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_poinsot_rolling/dynamic_routh_p2_poinsot_rolling.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_poinsot_rolling.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_poinsot_rolling.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_poinsot_rolling/player_dynamic_routh_p2_poinsot_rolling.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_polhode_quarter_period

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_polhode_quarter_period.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (14 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_polhode_quarter_period/dynamic_routh_p2_polhode_quarter_period.ipyn`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_polhode_quarter_period.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_polhode_quarter_period.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_polhode_quarter_period/player_dynamic_routh_p2_polhode_quarter_peric`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_principal_axes_in_space

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_principal_axes_in_space.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (15 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_principal_axes_in_space/dynamic_routh_p2_principal_axes_in_space.ip`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_principal_axes_in_space.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_principal_axes_in_space.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_principal_axes_in_space/player_dynamic_routh_p2_principal_axes_in_sp`
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dynamic_routh_p2_rolling_cones

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_rolling_cones.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (14 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_rolling_cones/dynamic_routh_p2_rolling_cones.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_rolling_cones.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_rolling_cones.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_rolling_cones/player_dynamic_routh_p2_rolling_cones.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_separatrix

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_separatrix.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (18 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_separatrix/dynamic_routh_p2_separatrix.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_separatrix.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_separatrix.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_separatrix/player_dynamic_routh_p2_separatrix.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_spin_stabilisation

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_spin_stabilisation.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (12 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_spin_stabilisation/dynamic_routh_p2_spin_stabilisation.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_spin_stabilisation.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_spin_stabilisation.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_spin_stabilisation/player_dynamic_routh_p2_spin_stabilisation.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_sylvester_time

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_sylvester_time.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (16 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_sylvester_time/dynamic_routh_p2_sylvester_time.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_sylvester_time.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_sylvester_time.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_sylvester_time/player_dynamic_routh_p2_sylvester_time.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_thin_rod

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_thin_rod.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (12 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_thin_rod/dynamic_routh_p2_thin_rod.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_thin_rod.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_thin_rod.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_thin_rod/player_dynamic_routh_p2_thin_rod.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_two_quadrics

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_two_quadrics.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (15 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_two_quadrics/dynamic_routh_p2_two_quadrics.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_two_quadrics.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_two_quadrics.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_two_quadrics/player_dynamic_routh_p2_two_quadrics.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_p2_uniaxal_precession

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_p2_uniaxal_precession.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (13 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_uniaxal_precession/dynamic_routh_p2_uniaxal_precession.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_p2_uniaxal_precession.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_p2_uniaxal_precession.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_p2_uniaxal_precession/player_dynamic_routh_p2_uniaxal_precession.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_routh_rectangle_diagonal

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/routh_rectangle_diagonal.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (17 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_rectangle_diagonal/dynamic_routh_rectangle_diagonal.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_routh_rectangle_diagonal.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/routh_rectangle_diagonal.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_routh_rectangle_diagonal/player_dynamic_routh_rectangle_diagonal.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_spin_up

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/spin_up.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (6 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_spin_up/dynamic_spin_up.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_spin_up.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/spin_up.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_spin_up/player_dynamic_spin_up.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_spinning_target

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/spinning_target.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (6 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_spinning_target/dynamic_spinning_target.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_spinning_target.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/spinning_target.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_spinning_target/player_dynamic_spinning_target.py run|gui|jump|capture out.png|data outdir`



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dynamic_static_anchor

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/static_anchor.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (7 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_static_anchor/dynamic_static_anchor.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_static_anchor.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/static_anchor.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_static_anchor/player_dynamic_static_anchor.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_thin_wall_toi

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/thin_wall_toi.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (9 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_thin_wall_toi/dynamic_thin_wall_toi.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_thin_wall_toi.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/thin_wall_toi.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_thin_wall_toi/player_dynamic_thin_wall_toi.py run|gui|jump|capture out.png|data outdir`



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dynamic_three_bodies

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/three_bodies.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (7 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_three_bodies/dynamic_three_bodies.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_three_bodies.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/three_bodies.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_three_bodies/player_dynamic_three_bodies.py`
`run|gui|jump|capture out.png|data outdir`



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dynamic_thrown_ball

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/thrown_ball.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (9 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_thrown_ball/dynamic_thrown_ball.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_thrown_ball.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/thrown_ball.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_thrown_ball/player_dynamic_thrown_ball.py`
`run|gui|jump|capture out.png|data outdir`



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127.0.0.1 refused to connect.

Try:

- Checking the connection
- [Checking the proxy and the firewall](#)

ERR_CONNECTION_REFUSED

Details

Reload

dynamic_tumbling_body

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/tumbling_body.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (13 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_tumbling_body/dynamic_tumbling_body.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_tumbling_body.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/tumbling_body.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_tumbling_body/player_dynamic_tumbling_body.py run|gui|jump|capture out.png|data outdir`



This site can't be reached

127.0.0.1 refused to connect.

Try:

- Checking the connection
- [Checking the proxy and the firewall](#)

ERR_CONNECTION_REFUSED

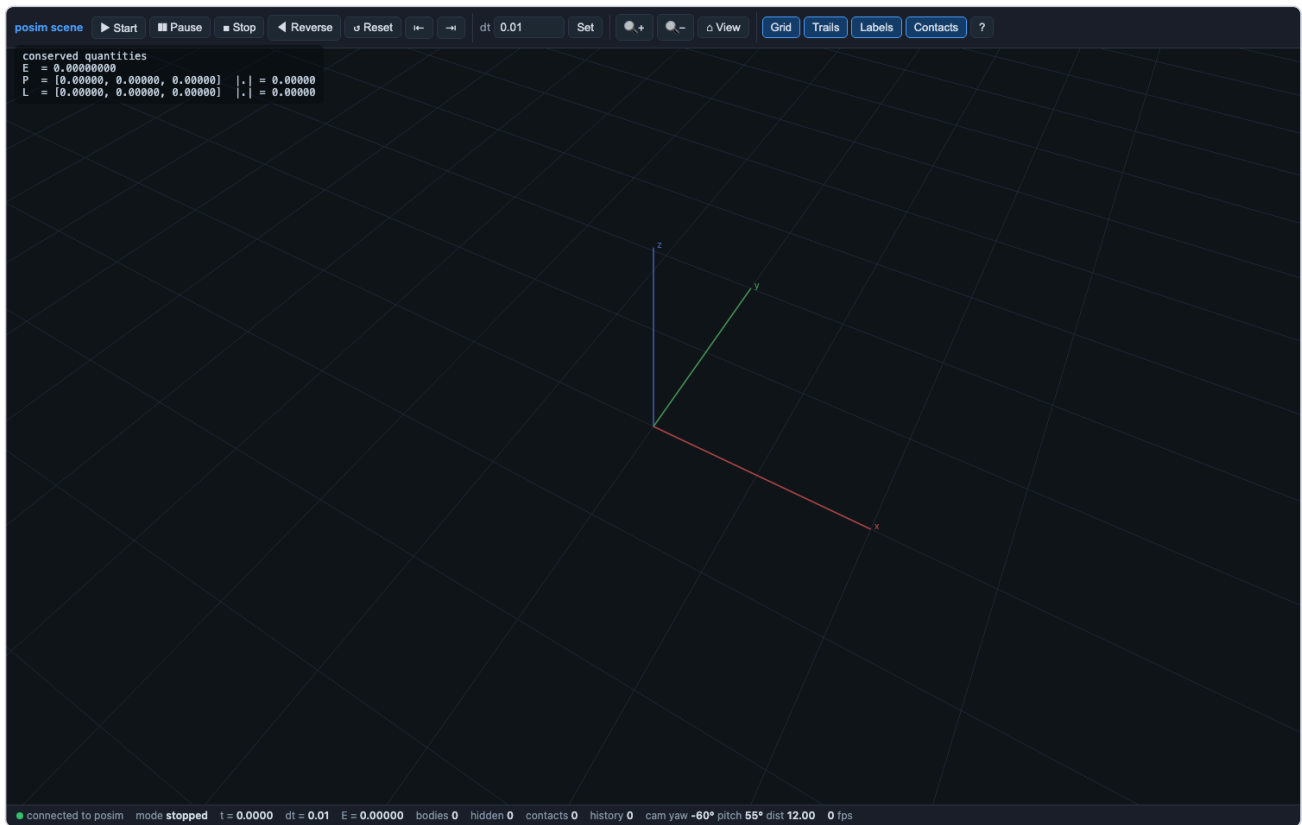
Details

Reload

dynamic_tunneling

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/tunneling.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (17 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_tunneling/dynamic_tunneling.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_tunneling.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/tunneling.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_tunneling/player_dynamic_tunneling.py`
`run|gui|jump|capture out.png|data outdir`



dynamic_two_dumbbells

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/two_dumbbells.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_two_dumbbells/dynamic_two_dumbbells.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_two_dumbbells.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/two_dumbbells.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_two_dumbbells/player_dynamic_two_dumbbells.py run|gui|jump|capture out.png|data outdir`



This site can't be reached

127.0.0.1 refused to connect.

Try:

- Checking the connection
- [Checking the proxy and the firewall](#)

ERR_CONNECTION_REFUSED

Details

Reload

dynamic_unequal_masses

This notebook is one half of a pair. The other half is the example file `dynamic_notebooks/unequal_masses.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (6 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_unequal_masses/dynamic_unequal_masses.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/dynamic_unequal_masses.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/dynamic_notebooks/unequal_masses.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/03_dynamics_and_routh/dynamic_unequal_masses/player_dynamic_unequal_masses.py run|gui|jump|capture out.png|data outdir`



This site can't be reached

127.0.0.1 refused to connect.

Try:

- Checking the connection
- [Checking the proxy and the firewall](#)

ERR_CONNECTION_REFUSED

Details

Reload

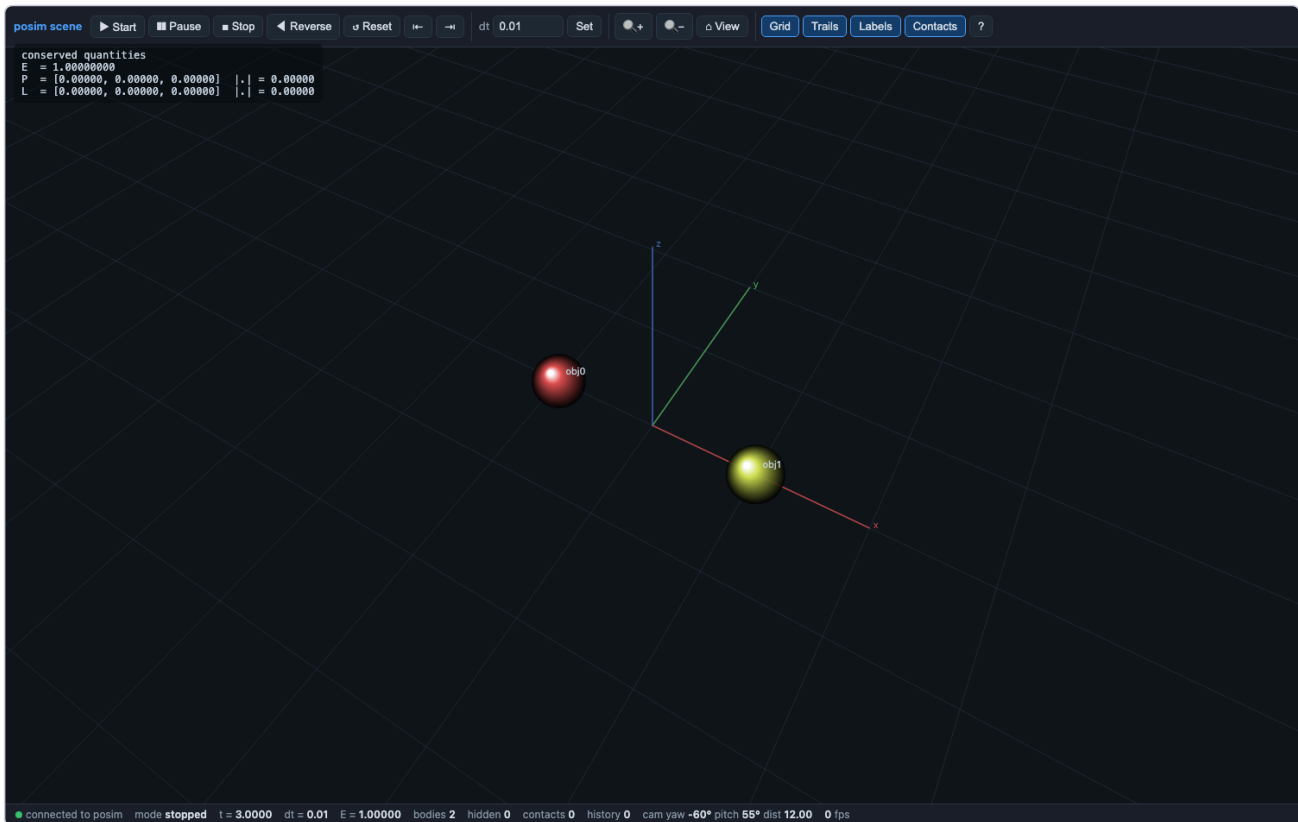
04_collisions

The collision-detection walkthroughs: elastic exchanges, restitution ladders, billiards, dumbbells, and the box-of-shapes stress test, all integrated by the pure-Rust engine.

collision_01_head_on_exchange

This notebook is one half of a pair. The other half is the example file `scripts/collisions/01_head_on_exchange.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

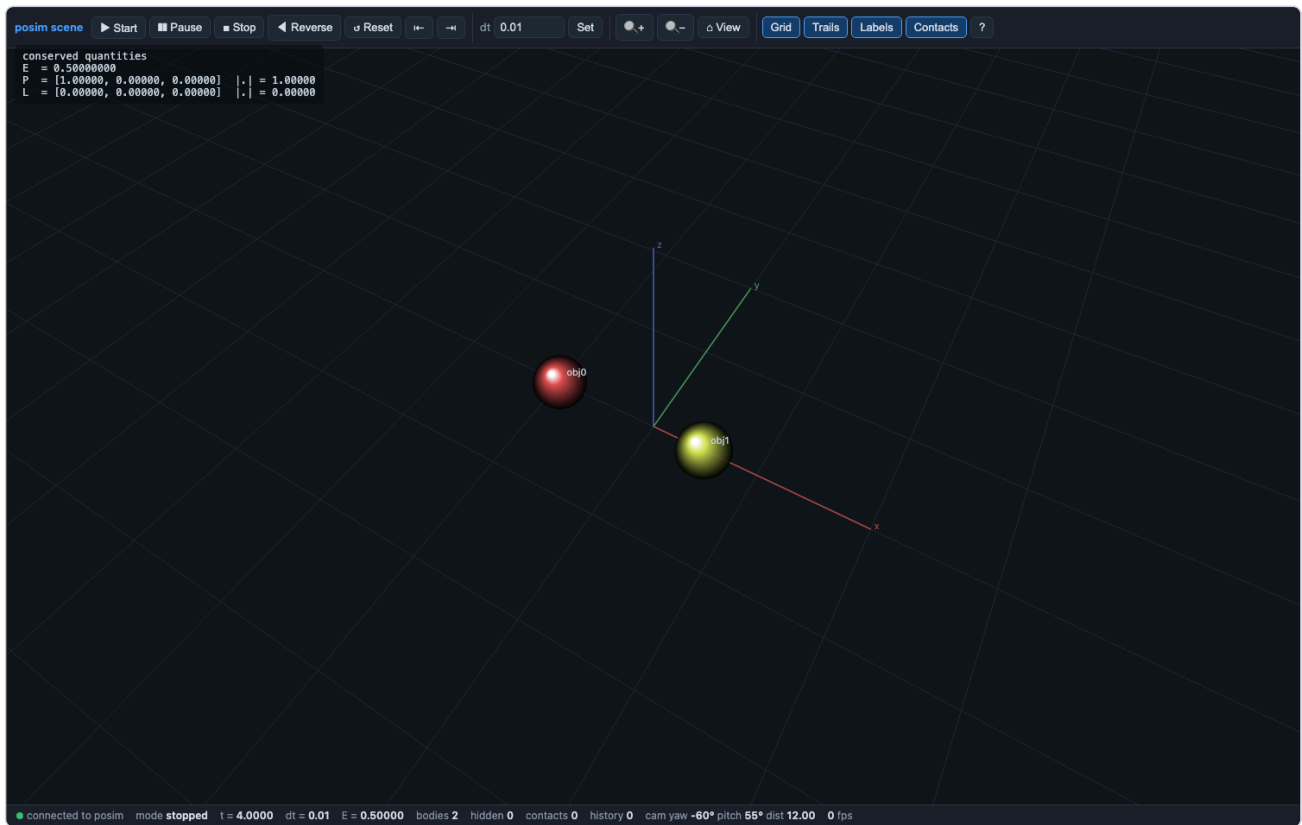
- **Executed:** PASS — every cell green (9 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/04_collisions/collision_01_head_on_exchange/collision_01_head_on_exchange.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_01_head_on_exchange.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/01_head_on_exchange.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/04_collisions/collision_01_head_on_exchange/player_collision_01_head_on_exchange.py`
`run|gui|jump|capture out.png|data outdir`



collision_02_unequal_masses

This notebook is one half of a pair. The other half is the example file `scripts/collisions/02_unequal_masses.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

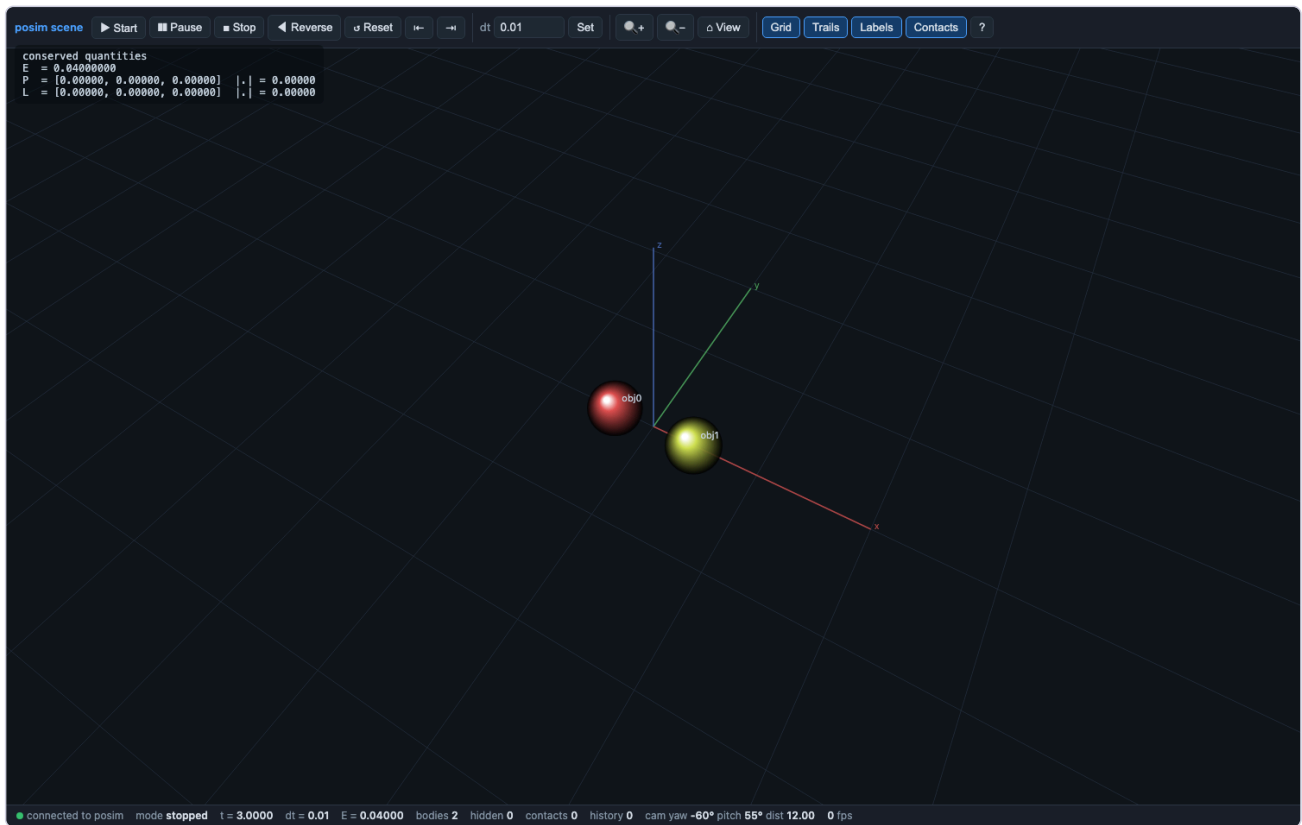
- **Executed:** PASS — every cell green (6 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/04_collisions/collision_02_unequal_masses/collision_02_unequal_masses.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_02_unequal_masses.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/02_unequal_masses.posim`
- **Player:** `python3` `SolveIt_Notebooks_for_rust/04_collisions/collision_02_unequal_masses/player_collision_02_unequal_masses.py`
`run|gui|jump|capture out.png|data outdir`



collision_03_restitution_ladder

This notebook is one half of a pair. The other half is the example file scripts/collisions/03_restitution_ladder.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

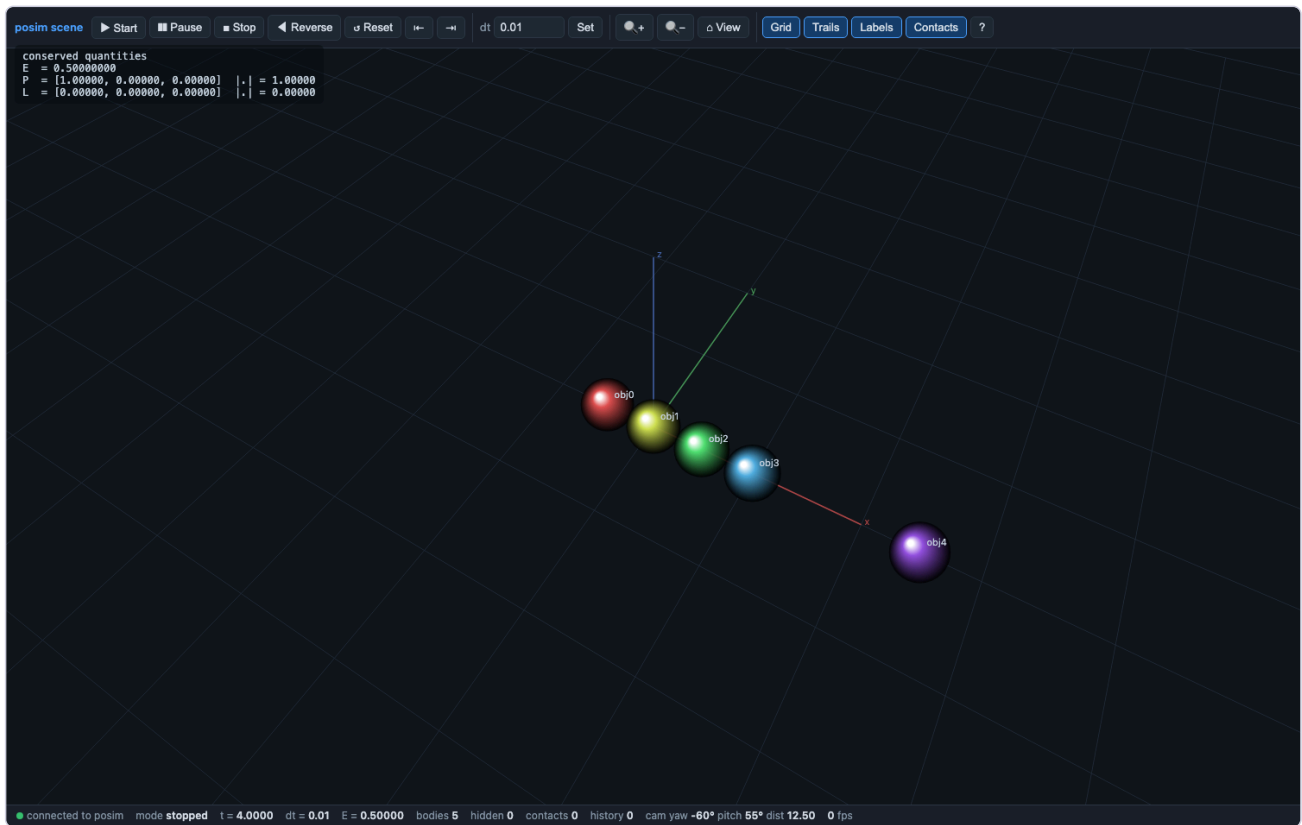
- **Executed:** PASS — every cell green (18 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/04_collisions/collision_03_restitution_ladder/collision_03_restitution_ladder.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_03_restitution_ladder.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/03_restitution_ladder.posim`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/04_collisions/collision_03_restitution_ladder/player_collision_03_restitution_ladder.py`
`run|gui|jump|capture out.png|data outdir`



collision_04_newtons_cradle

This notebook is one half of a pair. The other half is the example file `scripts/collisions/04_newtons_cradle.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

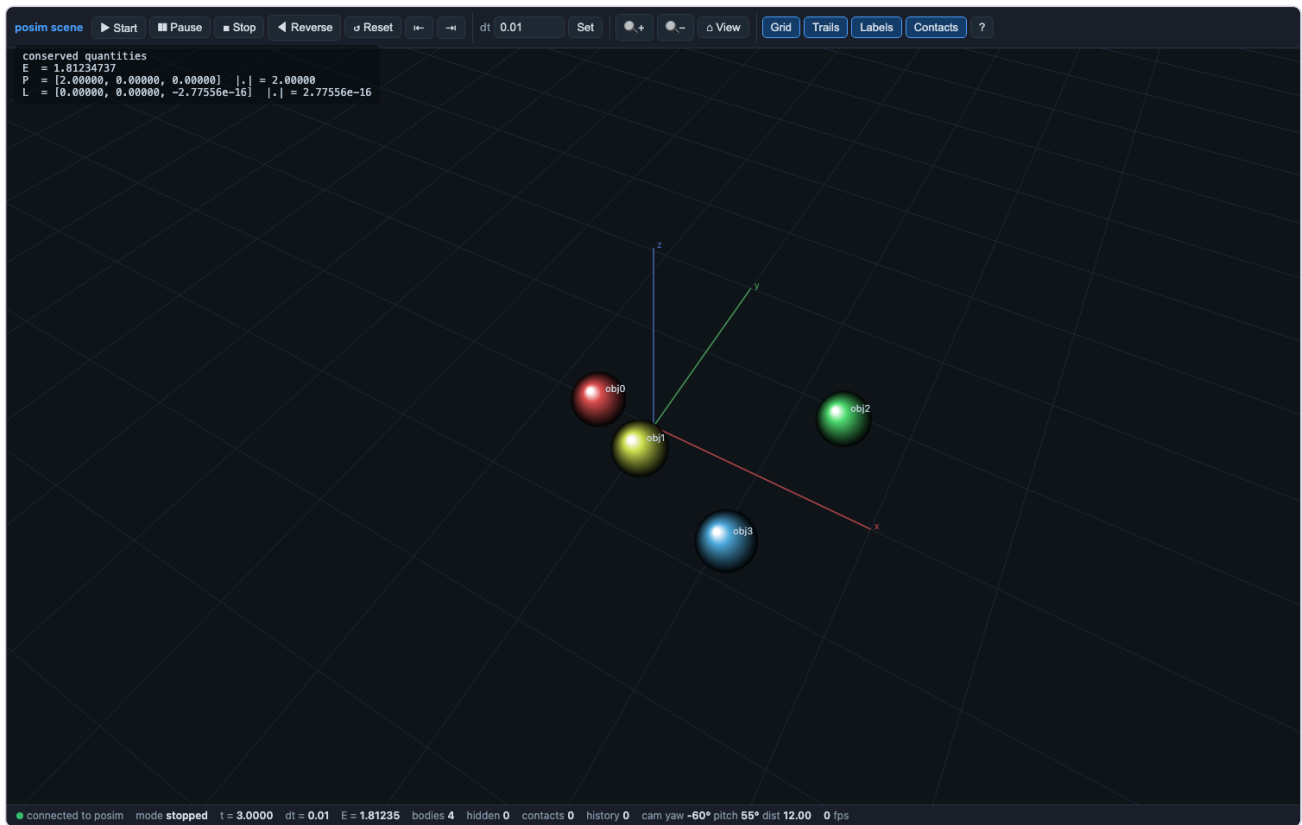
- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/04_collisions/collision_04_newtons_cradle/collision_04_newtons_cradle.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_04_newtons_cradle.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/04_newtons_cradle.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/04_collisions/collision_04_newtons_cradle/player_collision_04_newtons_cradle.py run|gui|jump|capture out.png|data outdir`



collision_05_billiard_break

This notebook is one half of a pair. The other half is the example file scripts/collisions/05_billiard_break.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

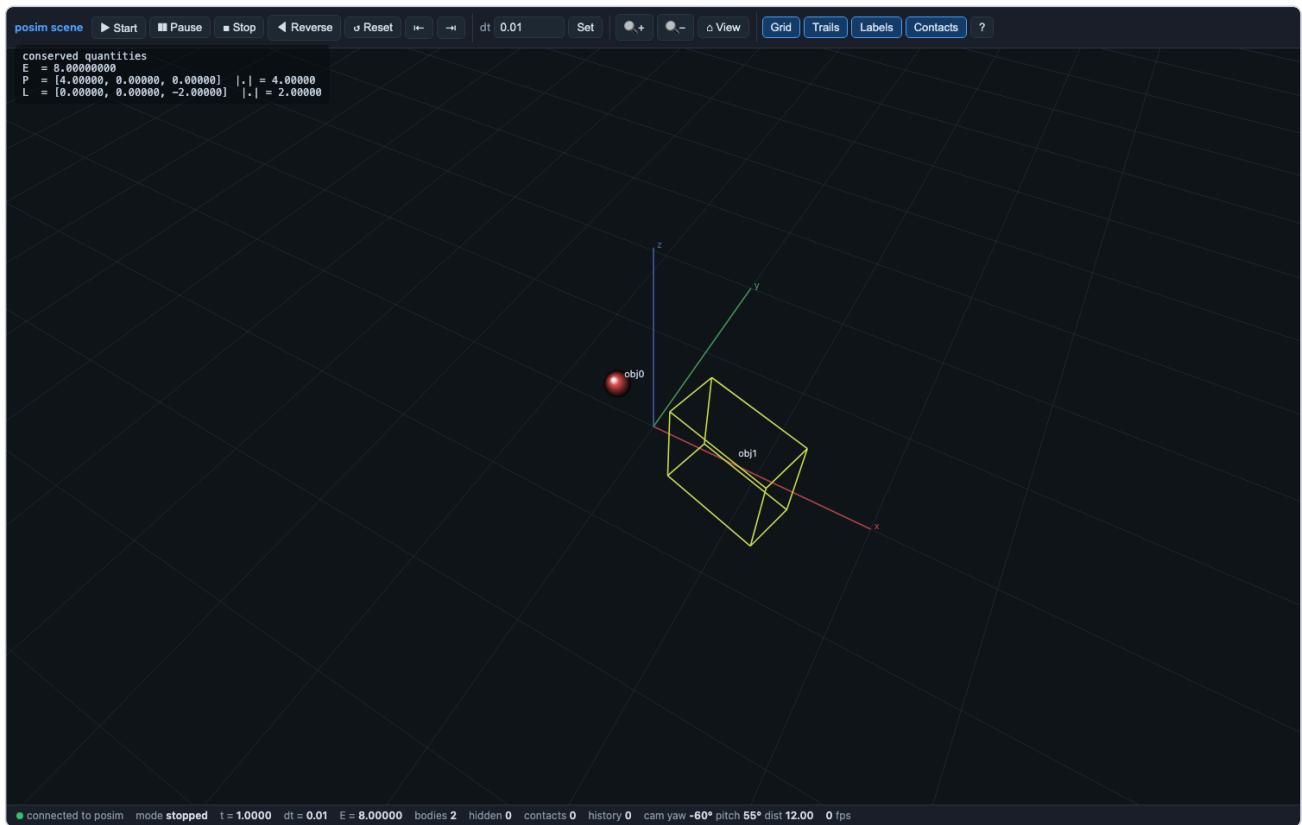
- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** SolveIt_Notebooks_for_rust/04_collisions/collision_05_billiard_break/collision_05_billiard_break.ipynb (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_05_billiard_break.ipynb
- **Paired scene/source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/05_billiard_break.posim
- **Player:** python3 SolveIt_Notebooks_for_rust/04_collisions/collision_05_billiard_break/player_collision_05_billiard_break.py
run|gui|jump|capture out.png|data outdir



collision_06_spin_up

This notebook is one half of a pair. The other half is the example file scripts/collisions/06_spin_up.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

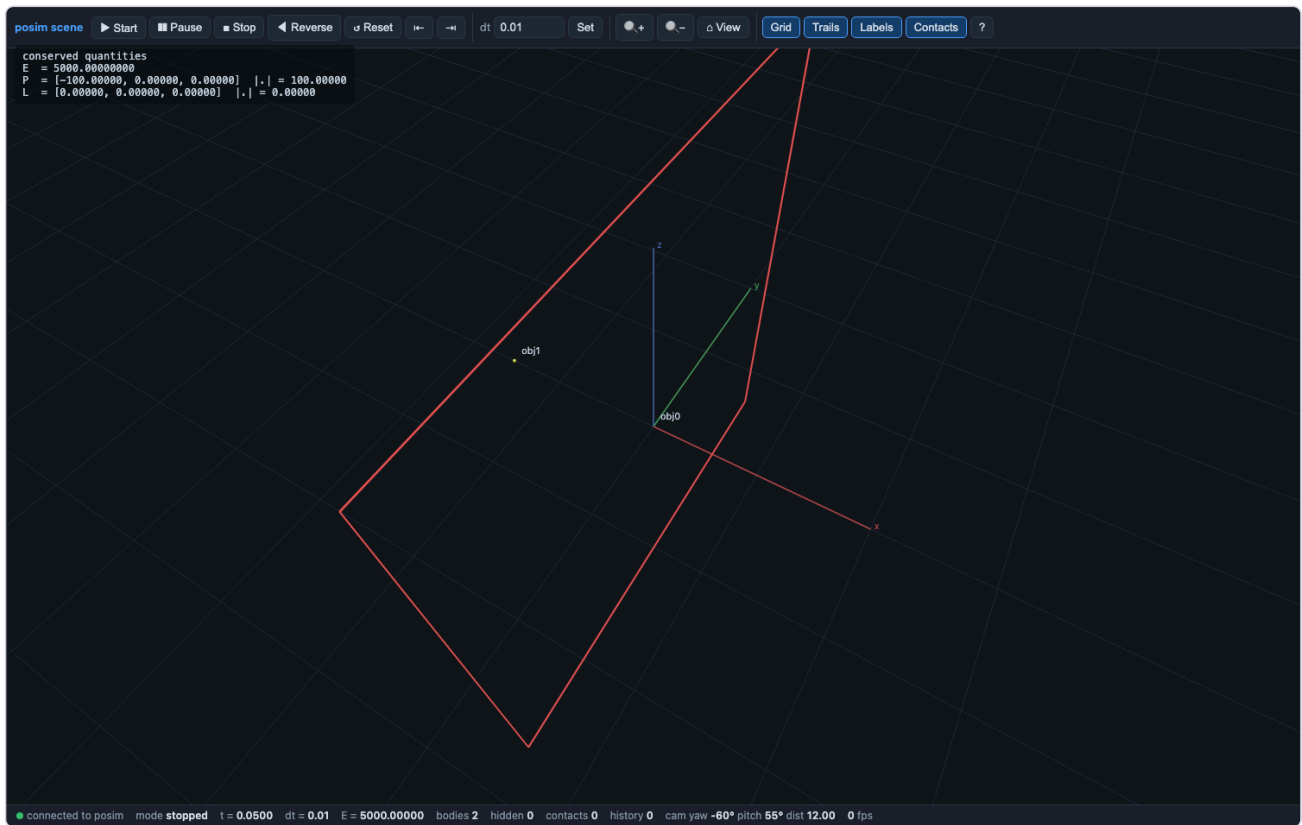
- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** SolveIt_Notebooks_for_rust/04_collisions/collision_06_spin_up/collision_06_spin_up.ipynb (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_06_spin_up.ipynb
- **Paired scene/source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/06_spin_up.posim
- **Player:** python3 SolveIt_Notebooks_for_rust/04_collisions/collision_06_spin_up/player_collision_06_spin_up.py
run|gui|jump|capture out.png|data outdir



collision_07_thin_wall_toi

This notebook is one half of a pair. The other half is the example file scripts/collisions/07_thin_wall_toi.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

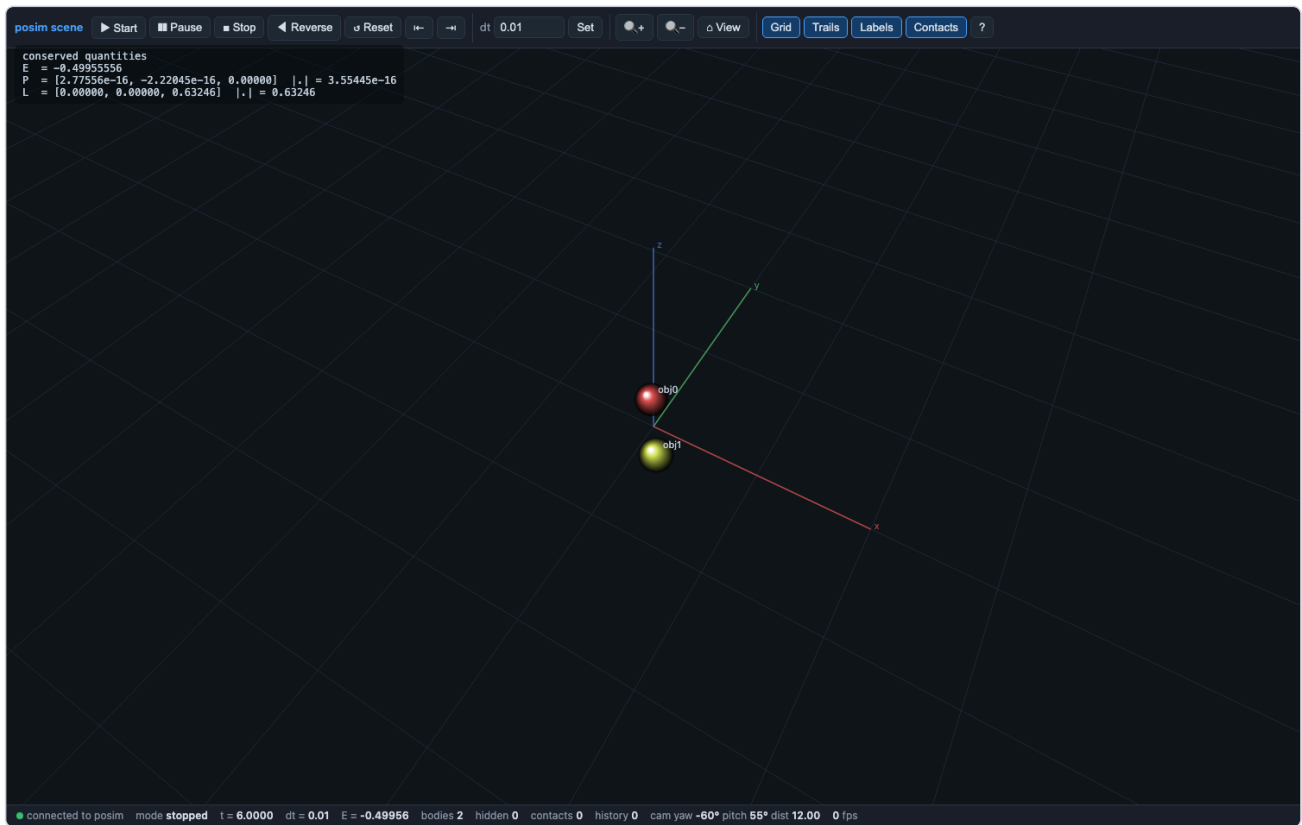
- **Executed:** PASS — every cell green (6 code cells)
- **Copy:** SolveIt_Notebooks_for_rust/04_collisions/collision_07_thin_wall_toi/collision_07_thin_wall_toi.ipynb (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_07_thin_wall_toi.ipynb
- **Paired scene/source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/07_thin_wall_toi.posim
- **Player:** python3 SolveIt_Notebooks_for_rust/04_collisions/collision_07_thin_wall_toi/player_collision_07_thin_wall_toi.py
run|gui|jump|capture out.png|data outdir



collision_08_colliding_binary

This notebook is one half of a pair. The other half is the example file scripts/collisions/08_colliding_binary.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

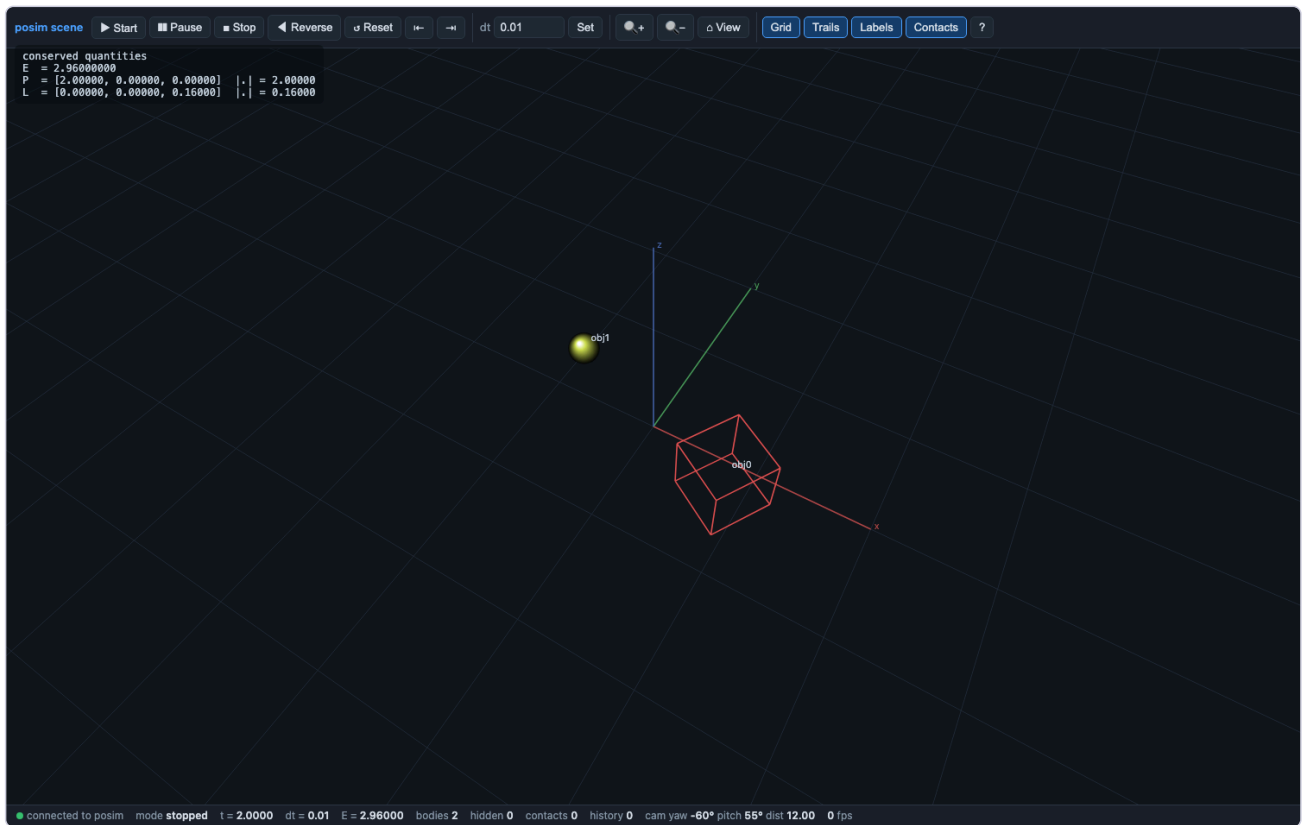
- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** SolveIt_Notebooks_for_rust/04_collisions/collision_08_colliding_binary/collision_08_colliding_binary.ipynb (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_08_colliding_binary.ipynb
- **Paired scene/source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/08_colliding_binary.posim
- **Player:** python3
SolveIt_Notebooks_for_rust/04_collisions/collision_08_colliding_binary/player_collision_08_colliding_binary.py
run|gui|jump|capture out.png|data outdir



collision_09_spinning_target

This notebook is one half of a pair. The other half is the example file scripts/collisions/09_spinning_target.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

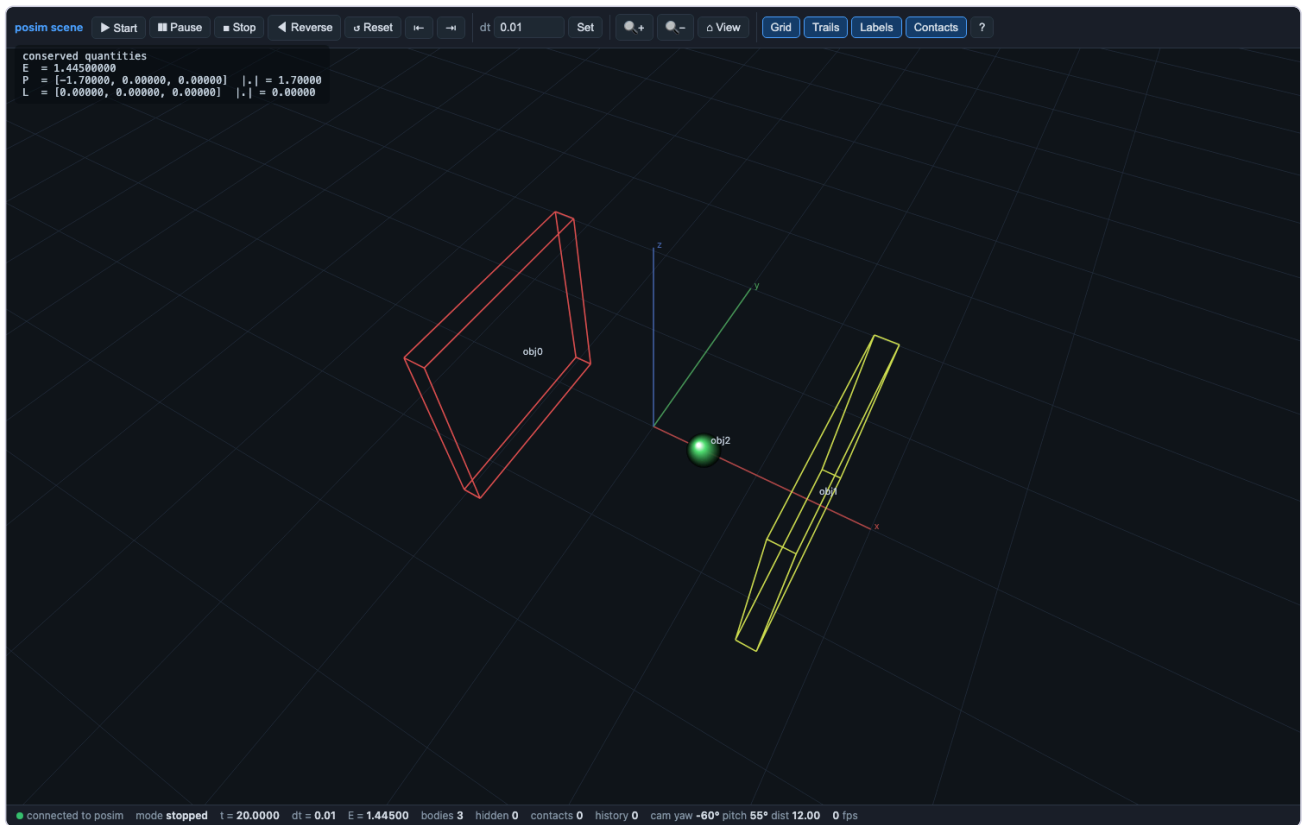
- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** SolveIt_Notebooks_for_rust/04_collisions/collision_09_spinning_target/collision_09_spinning_target.ipynb (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_09_spinning_target.ipynb
- **Paired scene/source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/09_spinning_target.posim
- **Player:** python3 SolveIt_Notebooks_for_rust/04_collisions/collision_09_spinning_target/player_collision_09_spinning_target.py
run|gui|jump|capture out.png|data outdir



collision_10_billiard_box

This notebook is one half of a pair. The other half is the example file `scripts/collisions/10_billiard_box.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

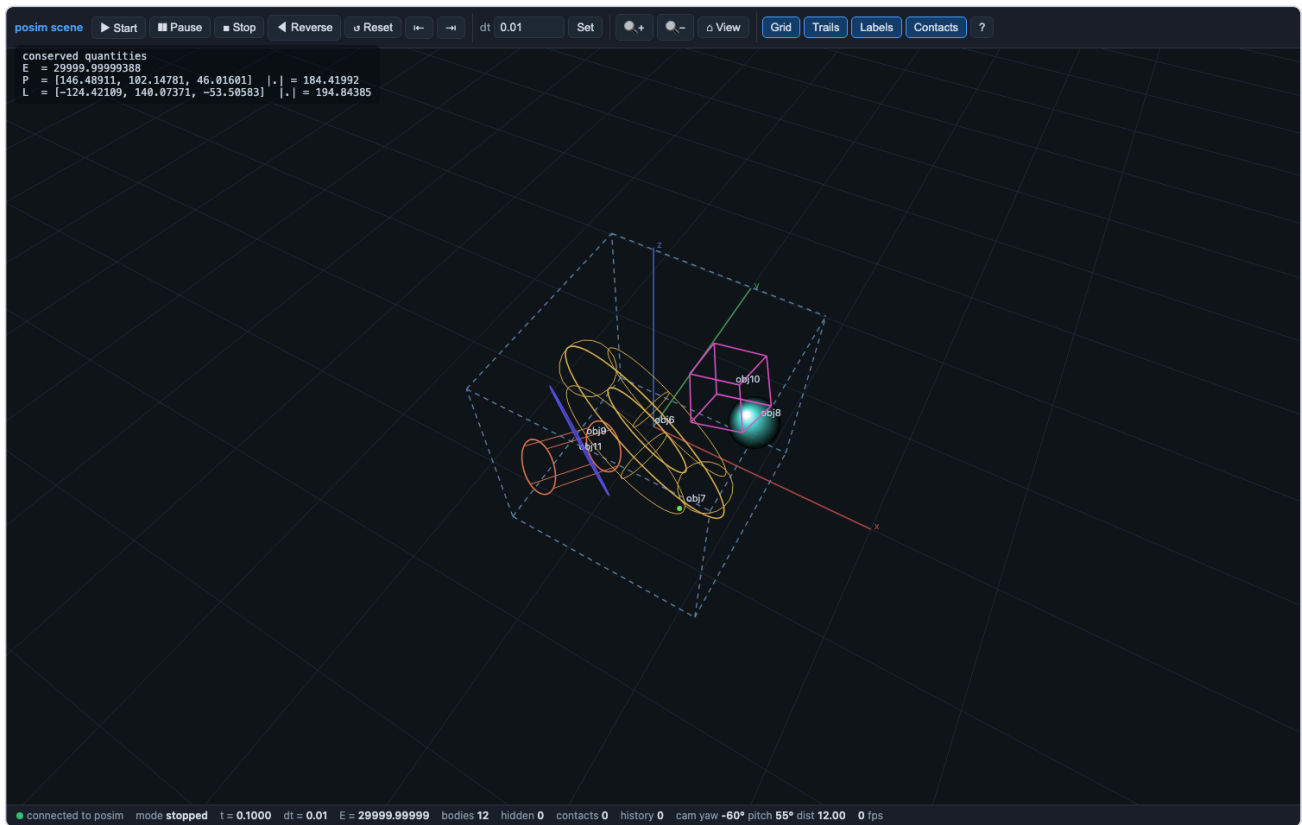
- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/04_collisions/collision_10_billiard_box/collision_10_billiard_box.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_10_billiard_box.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/10_billiard_box.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/04_collisions/collision_10_billiard_box/player_collision_10_billiard_box.py`
`run|gui|jump|capture out.png|data outdir`



collision_11_box_of_shapes

This notebook is one half of a pair. The other half is the example file scripts/collisions/11_box_of_shapes.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

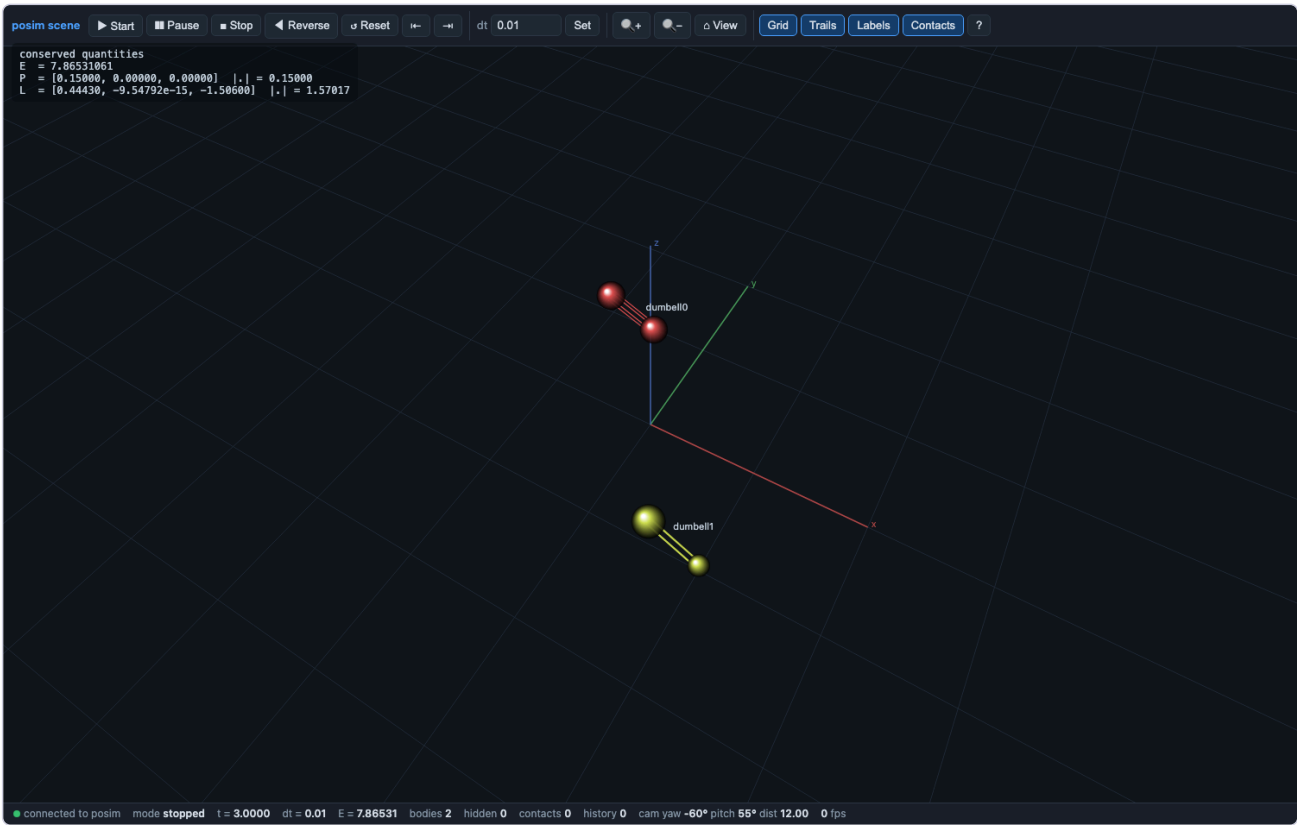
- **Executed:** PASS — every cell green (10 code cells)
- **Copy:** SolveIt_Notebooks_for_rust/04_collisions/collision_11_box_of_shapes/collision_11_box_of_shapes.ipynb (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_11_box_of_shapes.ipynb
- **Paired scene/source:** rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/11_box_of_shapes.posim
- **Player:** python3 SolveIt_Notebooks_for_rust/04_collisions/collision_11_box_of_shapes/player_collision_11_box_of_shapes.py
run|gui|jump|capture out.png|data outdir



collision_12_two_dumbbells

This notebook is one half of a pair. The other half is the example file `scripts/collisions/12_two_dumbbells.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (9 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/04_collisions/collision_12_two_dumbbells/collision_12_two_dumbbells.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/collision_12_two_dumbbells.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/scripts/collisions/12_two_dumbbells.posim`
- **Player:** `python3 SolveIt_Notebooks_for_rust/04_collisions/collision_12_two_dumbbells/player_collision_12_two_dumbbells.py`
`run|gui|jump|capture out.png|data outdir`



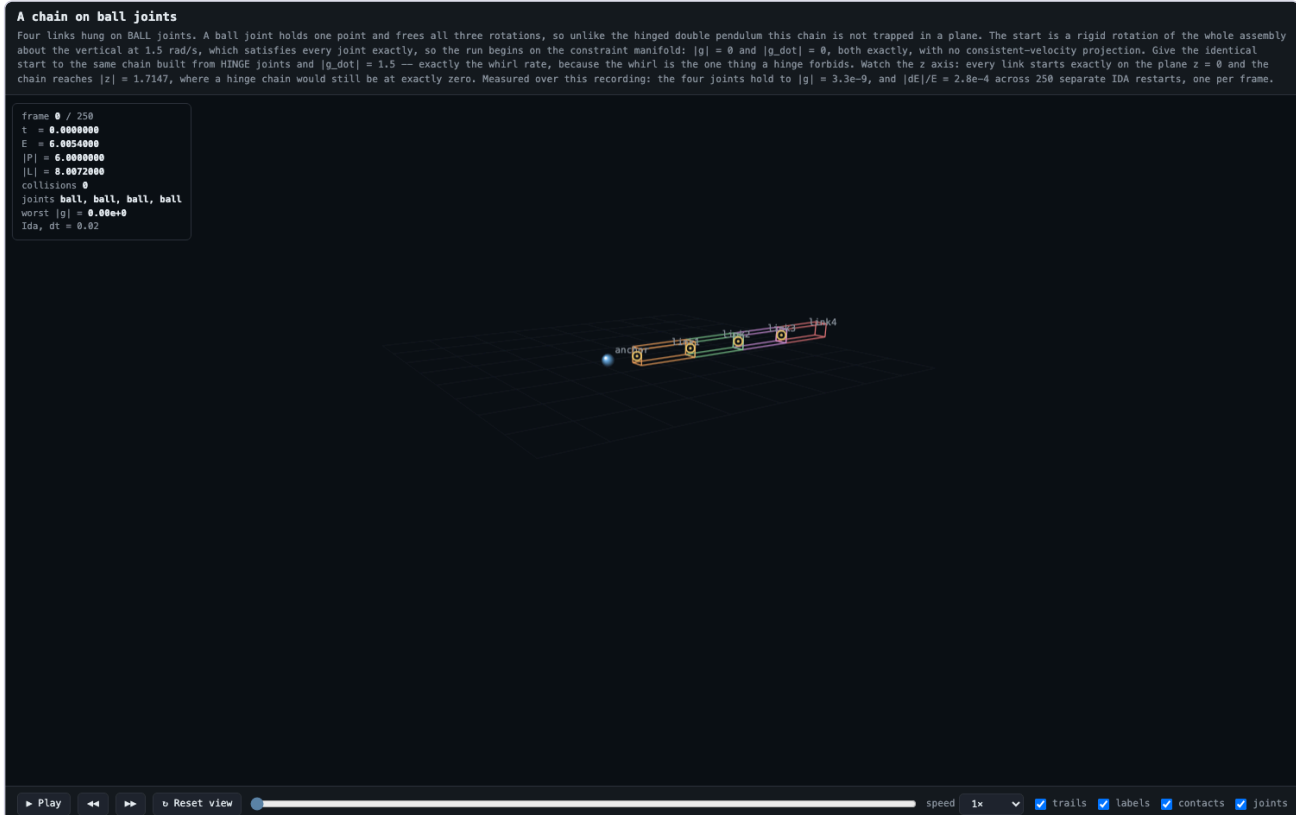
05_mechanism_videos

The thirteen recorded mechanism scenes: each notebook pairs with a recorded browser movie page in videos/ AND a live GUI server in gui/ that integrates the same scene in real time behind canvas controls with live physics readouts.

video_ball_joint_chain

This notebook is one half of a pair. The other half is the example file videos/scenes/ball_joint_chain.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

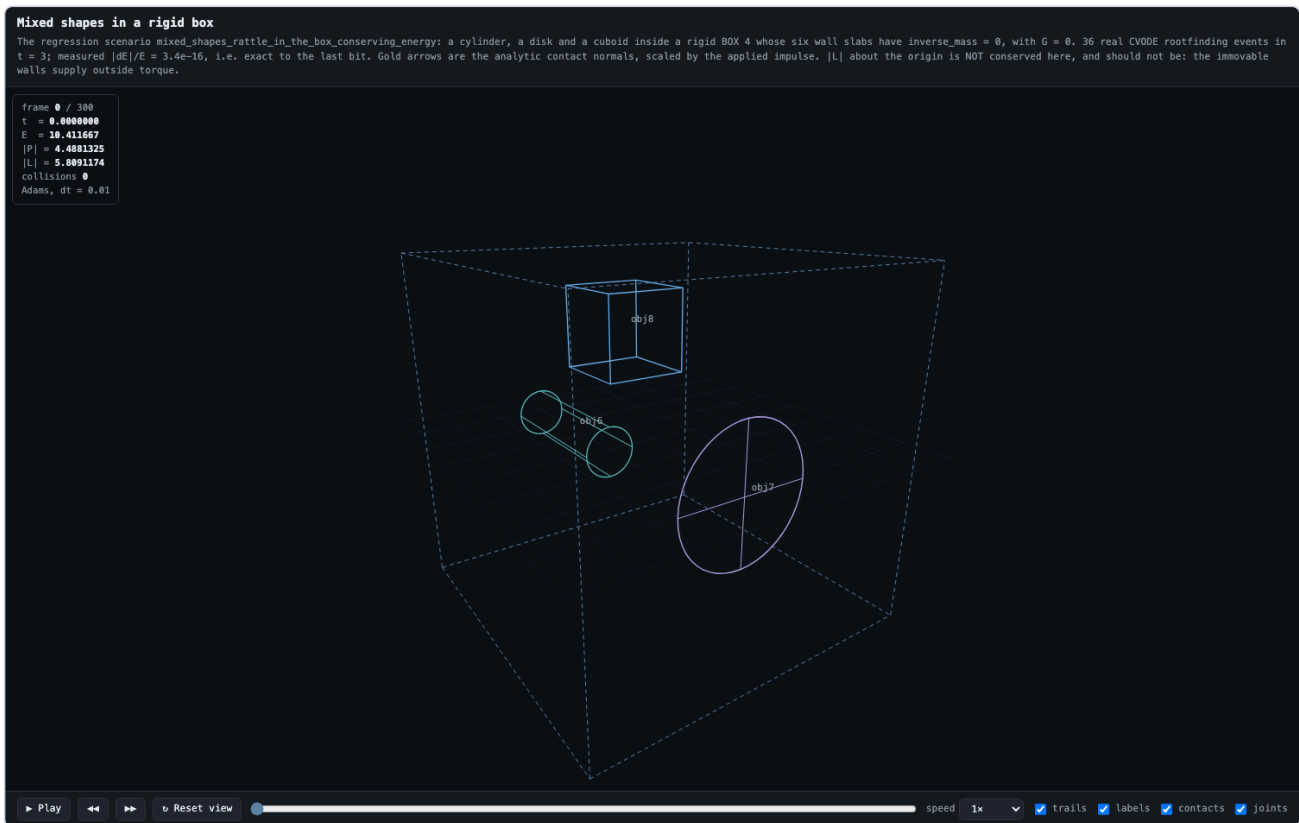
- **Executed:** PASS — every cell green (7 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_ball_joint_chain/video_ball_joint_chain.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_ball_joint_chain.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/ball_joint_chain.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/ball_joint_chain.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/ball_joint_chain/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_ball_joint_chain/player_video_ball_joint_chain.py`
`run|gui|jump|capture out.png|data outdir`



video_box_of_shapes

This notebook is one half of a pair. The other half is the example file videos/scenes/box_of_shapes.posim in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (5 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_box_of_shapes/video_box_of_shapes.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_box_of_shapes.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/box_of_shapes.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/box_of_shapes.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/box_of_shapes/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_box_of_shapes/player_video_box_of_shapes.py`
`run|gui|jump|capture out.png|data outdir`



video_cardan_compass

This notebook is one half of a pair. The other half is the example file `videos/scenes/cardan_compass.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_cardan_compass/video_cardan_compass.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_cardan_compass.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/cardan_compass.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/cardan_compass.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/cardan_compass/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_cardan_compass/player_video_cardan_compass.py`
`run|gui|jump|capture out.png|data outdir`

A compass in a Cardan suspension

Two HINGE joints on two perpendicular horizontal axes: 10 rows on 12 freedoms, leaving the roll and pitch the suspension exists to give. Both rings lie FLAT, because a gimbal ring pivots about a DIAMETER — a line in its own plane — so its plane must contain both hinge axes. The difference from the gyroscope gimbal is one number. There every centre of mass sat ON the pivot, the lever arm was zero and gravity did nothing; here the bowl is PENDULOUS, its centre of mass hanging $d = 0.12$ below the pivot, and that is the whole mechanism. A ship's compass is not held level by its bearings — it is held level by its own weight, and the bearings only get out of the way. So each axis is a physical pendulum with a period you can write down, $T = 2\pi \sqrt{I/(Mgd)}$: 1.876597 s in pitch, where only the bowl swings, and 2.307339 s in roll, where ring and bowl swing together. Measured off this recording: 1.883426 s and 2.313853 s. The $2.6\text{e-}3$ excess is not solver error but finite amplitude — halve the kick and it falls by 4.81 and 4.02 , the signature of a theta-squared term. Excited on one axis alone the attribution is exact: the pitch period comes out $4.99\text{e-}4$ long against a predicted $\theta^2/16 = 5.02\text{e-}4$. The two hinges hold to $|g| = 3.4\text{e-}9$; the card never tilts more than 7.34 degrees off level and the ring swings 5.26 degrees about its hinge.

```
frame 0 / 320
t = 0.000000
E = -2.1005193
|P| = 0.00000000
|L| = 0.095341327
collisions 0
joints hinge, hinge
worst |g| = 8.33e-17
ids, dt = 0.025
```



► Play ◀◀ ▶▶ ⚙ Reset view speed 1x ▾ ☒ trails ☒ labels ☒ contacts ☒ joints

video_cardan_gear

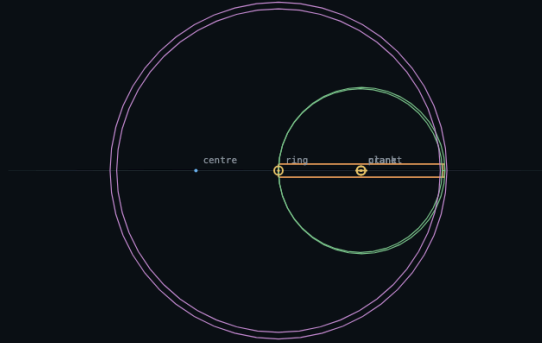
This notebook is one half of a pair. The other half is the example file `videos/scenes/cardan_gear.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (10 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_cardan_gear/video_cardan_gear.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_cardan_gear.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/cardan_gear.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/cardan_gear.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/cardan_gear/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_cardan_gear/player_video_cardan_gear.py`
`run|gui|jump|capture out.png|data outdir`

Cardan gears, and a straight line

A wheel of radius r inside a ring of radius $2r$: every point on the wheel's rim travels along a straight line, a diameter of the ring. The hypocycloid of ratio 2 is degenerate — it is not an approximation to a straight line, it IS one. The rolling is a GEAR joint: one row, holding the two bodies' turns about a shared axis in proportion, ratio 1 because the wheel turns once backwards for each forward turn of the carrier. It stacks on the bearing rather than replacing it, so the mechanism is HINGE 5 + GEAR 1 = 11 rows on 12 freedoms, leaving the one freedom a gear train has. The ratio must be rational: the honest constraint is on ACCUMULATED angle, which a quaternion cannot report, so the row is $\sin(q\theta_{\text{theta}_1} + p\theta_{\text{theta}_2})$ with the angles wrapped — which is faithful only because p and q are whole numbers, and an irrational ratio is refused rather than rounded. What the constraint buys, measured on this scene: impose the same relation as an initial condition instead and the rim leaves the line by $1.795e-3$ against the gear's $1.059e-8$; then lean on the wheel with a torque and the imposed version collapses to $7.468e-1$ while the gear holds $1.152e-8$. The outer ring is drawn for reference and nothing is attached to it — the gear row, not any contact, does the rolling.

```
frame 0 / 315
t = 0.0000000
E = 0.23750000
|P| = 0.65000000
|L| = 0.22516000
collisions 0
joints hinge, hinge, gear
worst |g| = 0.00e+0
ids, dt = 0.02
```

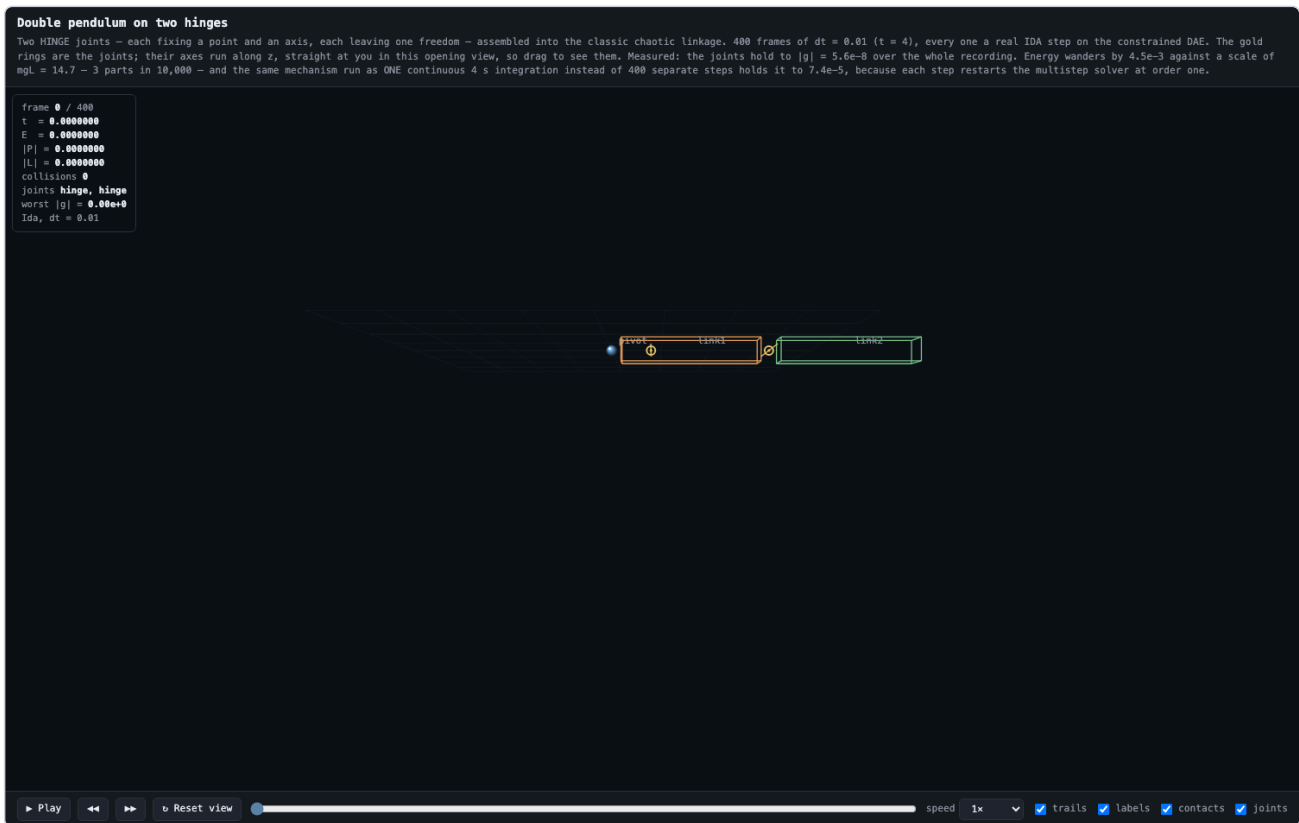


► Play ◀◀ ▶▶ ◂ Reset view speed 1x ☒ trails ☒ labels ☒ contacts ☒ joints

video_double_pendulum_hinges

This notebook is one half of a pair. The other half is the example file `videos/scenes/double_pendulum_hinges.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (6 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_double_pendulum_hinges/video_double_pendulum_hinges.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_double_pendulum_hinges.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/double_pendulum_hinges.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/double_pendulum_hinges.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/double_pendulum_hinges/server.py`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/05_mechanism_videos/video_double_pendulum_hinges/player_video_double_pendulum_hinges.py`
`run|gui|jump|capture out.png|data outdir`



video_gyroscope_gimbal

This notebook is one half of a pair. The other half is the example file `videos/scenes/gyroscope_gimbal.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

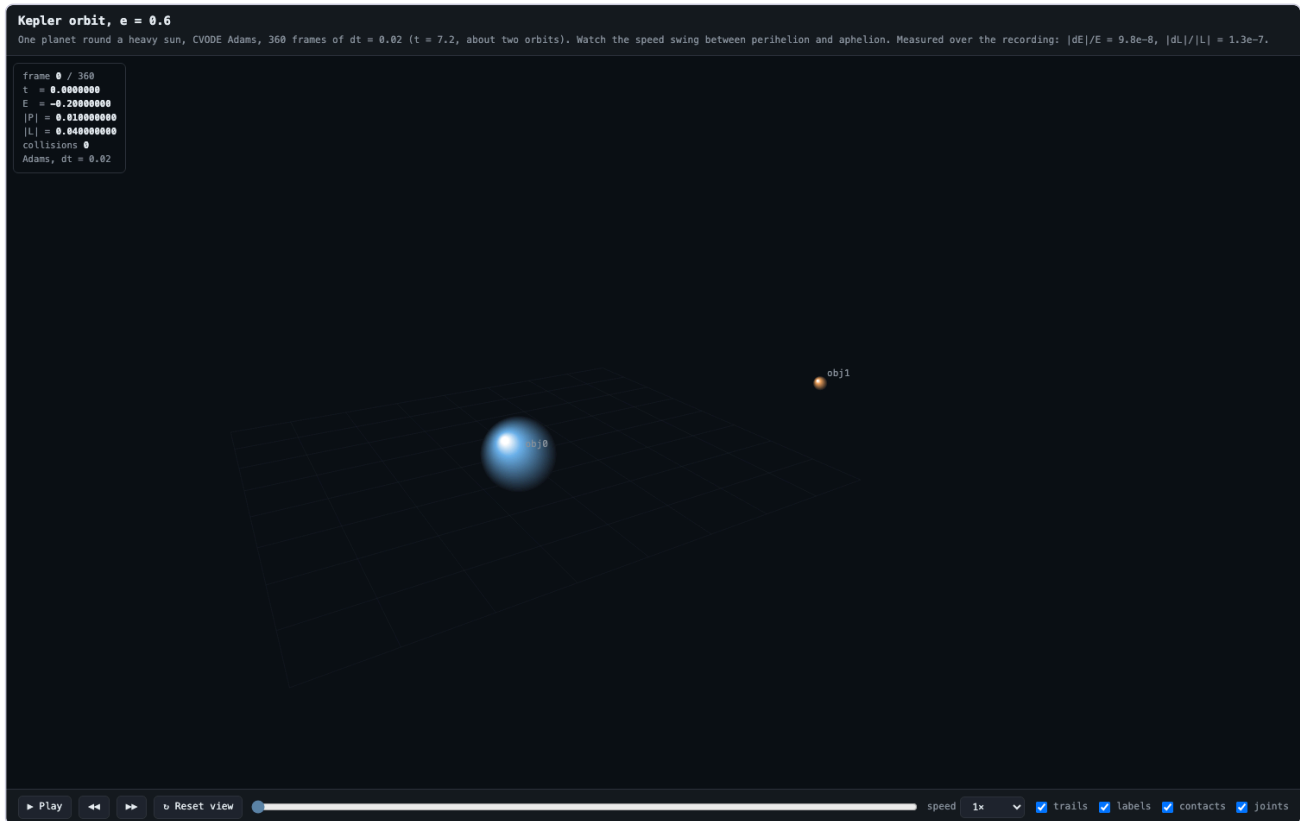
- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_gyroscope_gimbal/video_gyroscope_gimbal.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_gyroscope_gimbal.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/gyroscope_gimbal.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/gyroscope_gimbal.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/gyroscope_gimbal/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_gyroscope_gimbal/player_video_gyroscope_gimbal.py`
`run|gui|jump|capture out.png|data outdir`



video_kepler_ellipse

This notebook is one half of a pair. The other half is the example file `videos/scenes/kepler_ellipse.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (5 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_kepler_ellipse/video_kepler_ellipse.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_kepler_ellipse.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/kepler_ellipse.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/kepler_ellipse.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/kepler_ellipse/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_kepler_ellipse/player_video_kepler_ellipse.py`
`run|gui|jump|capture out.png|data outdir`



video_piston_crankshaft

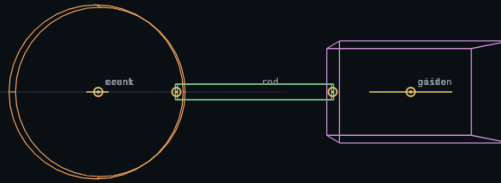
This notebook is one half of a pair. The other half is the example file `videos/scenes/piston_crankshaft.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (10 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_piston_crankshaft/video_piston_crankshaft.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_piston_crankshaft.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/piston_crankshaft.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/piston_crankshaft.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/piston_crankshaft/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_piston_crankshaft/player_video_piston_crankshaft.py`
`run|gui|jump|capture out.png|data outdir`

A piston and crankshaft

The slider-crank: mount—HINGE—crank, crank—BALL—rod, rod—BALL—piston, guide—PRISMATIC—piston. 5 + 3 + 3 + 5 = 16 rows on 18 freedoms. Every pin in a real engine is a hinge, and building it that way gives 20 rows on 18 — over-constrained by two, which IDA will not integrate, because a planar linkage made of spatial revolute has them all insisting on the same plane. The remedy is the one real multibody models use: spherical ends on the connecting rod. That leaves two freedoms, the crank angle and the rod's passive spin about its own length, which nothing torques. The geometry is not arbitrary either — the midpoint pivot rule puts the rod's centre at 2a while a rod from crank pin to wrist pin centres at a + L/2, and those agree only when L = 2a, which is why this crank throws 0.5 and this rod is 1.0. Measuring the wrist pin from the crankshaft axis, $x = a \cos(th) + \sqrt{a^2 - r^2} \sin^2(th)$ exactly: the recording sweeps 0.500000 to 1.500000, which is L-a to L+a, and tracks the formula to 8.4e-8 over the 1.59 revolutions it makes. The rod's passive freedom really is passive: its spin about its own length is 0.000 degrees over the whole run. The crank does not turn at a constant rate and should not — nothing drives it, so it swaps inertia with the rod and piston through the cycle. The closed form is a statement about the linkage, not the timing, which is what makes it worth checking against a free-running mechanism.

```
frame 0 / 300
t = 0.0000000
E = 0.67183333
|P| = 0.20000000
|L| = 0.77333333
collisions 0
joints hinge, ball, ball, prismatic
worst |g| = 0.00e+0
ids, dt = 0.02
```



► Play ◀◀ ▶▶ ↺ Reset view speed 1x ☒ trails ☒ labels ☒ contacts ☒ joints

video_rack_and_pinion

This notebook is one half of a pair. The other half is the example file `videos/scenes/rack_and_pinion.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_rack_and_pinion/video_rack_and_pinion.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_rack_and_pinion.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/rack_and_pinion.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/rack_and_pinion.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/rack_and_pinion/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_rack_and_pinion/player_video_rack_and_pinion.py`
`run|gui|jump|capture out.png|data outdir`

A rack and pinion drive

A falling weight turning a flywheel, and every joint in it was added for this: mount—HINGE—pinion, guide—PRISMATIC—bar, pinion=RACK=bar. 5 + 5 + 1 = 11 rows on 12 freedoms, leaving one. The RACK ties the pinion's turn to the bar's travel; the PRISMATIC is the guide a rack actually runs in — drop it and the reaction torque twists the bar 24.3 degrees off square and shoves it 0.68 off its line, while with it the twist is 2.6e-22 degrees and the sideways stray is exactly 0.0. The drive has one degree of freedom, so one equation: $a = m g / (m + I/r^2)$, the flywheel acting not with its mass but with its inertia referred through the pitch radius. For a solid disc $I/r^2 = M/2$, independent of the radius entirely, so a pinion twice the rack's mass makes that term exactly m and the rack falls at $a = g/2$ — exactly half gravity, whatever the pitch radius, because the flywheel it is winding up is, referred through the teeth, exactly as heavy as itself. The pitch radius here is 0.6, and it was 0.4 in an earlier cut with the fall unchanged — the radius-independence demonstrated rather than asserted. Measured against $-a t^2/2$ over four seconds and a 1.6 drop: 5.35e-4 across 200 cold restarts, 3.74e-5 for the same four seconds run continuously.

```
frame 0 / 200
t = 0.0000000
E = 0.0000000
|P| = 0.0000000
|L| = 0.0000000
collisions 0
joints hinge, prismatic, rack
worst |g| = 0.00e+0
ids, dt = 0.02
```



► Play ◀▶ ◂ Reset view speed 1x ☒ trails ☒ labels ☒ contacts ☒ joints

video_rod_pendulum_chain

This notebook is one half of a pair. The other half is the example file `videos/scenes/rod_pendulum_chain.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_rod_pendulum_chain/video_rod_pendulum_chain.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_rod_pendulum_chain.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/rod_pendulum_chain.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/rod_pendulum_chain.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/rod_pendulum_chain/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_rod_pendulum_chain/player_video_rod_pendulum_chain.py`
`run|gui|jump|capture out.png|data outdir`

Four bobs on four rods

A CONSTRAINT is the smallest joint there is: one row, holding one scalar fact, $g = |d| - L$. Four rods on four free bobs is 4 rows on 24 freedoms, against 3 rows per BALL, 4 per UNIVERSAL and 5 per HINGE. It is also the best conditioned joint — one well-scaled scalar equation, with none of the index-2 orientation coupling that forces a tolerance floor on the other three — so run this chain as ONE integration at the default $1e-10/1e-12$ and the rods hold to $|g| = 5.4e-15$, which is roundoff. This recording is not one integration, though. It is 250 cold restarts, one per frame, and the tolerance a continuous run sustains is not the tolerance a restart sustains: at the default these same four rods fail on the SECOND restart. So the scene asks for $1e-6/1e-8$ by hand — exactly the floor an orientation joint gets automatically and never has to request. Measured over this recording: the four rods hold to $|g| = 7.8e-8$. Released from rest, so the start is on the constraint manifold with nothing to project.

```
frame 0 / 250
t = 0.0000000
E = -3.2211075
|P| = 0.0000000
|L| = 0.0000000
collisions 0
joints rod, rod, rod, rod
worst |g| = 0.00e+0
Ida, dt = 0.02
```



► Play ◀◀ ▶▶ ◂ Reset view speed 1x ☒ trails ☒ labels ☒ contacts ☒ joints

video_spinning_top

This notebook is one half of a pair. The other half is the example file `videos/scenes/spinning_top.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_spinning_top/video_spinning_top.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_spinning_top.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/spinning_top.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/spinning_top.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/spinning_top/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_spinning_top/player_video_spinning_top.py`
`run|gui|jump|capture out.png|data outdir`

A spinning top, precession

A top held at its tip by a BALL joint — one point held, all three rotations free, which is exactly what the tip of a top is. A symmetric top spinning at ω_3 about its own axis, its centre of mass a distance r from the pivot, precesses at $\Omega = Mgr/(I_3 \omega_3)$. That is usually the fast-top APPROXIMATION, but this top is mounted with its axis horizontal, and at 90 degrees the correction term carries a $\cos(\theta)$ that is exactly zero — so the formula is exact here. With $M = 1$, $r = 0.6$, $g = 3$ and $I_3 = 0.0882$, spinning at 20 rad/s, it predicts 1.0204081632653061 rad/s and a period of 6.1575 s. Measured over this recording: 1.028440 rad/s, three parts in 100,000, with the axis staying horizontal to $4.3e-5$ — it does not nutate at all, because it was started ON the steady solution, which takes the precession velocity as well as the spin. Released with only its spin it misses the formula by 7.8%. The joint holds to $|g| = 4.3e-7$.

```
frame 0 / 310
t = 0.0000000
E = 17.851066
|P| = 0.61224400
|L| = 1.8118326
collisions 0
joints ball
worst |g| = 0.00e+0
Ida, dt = 0.02
```

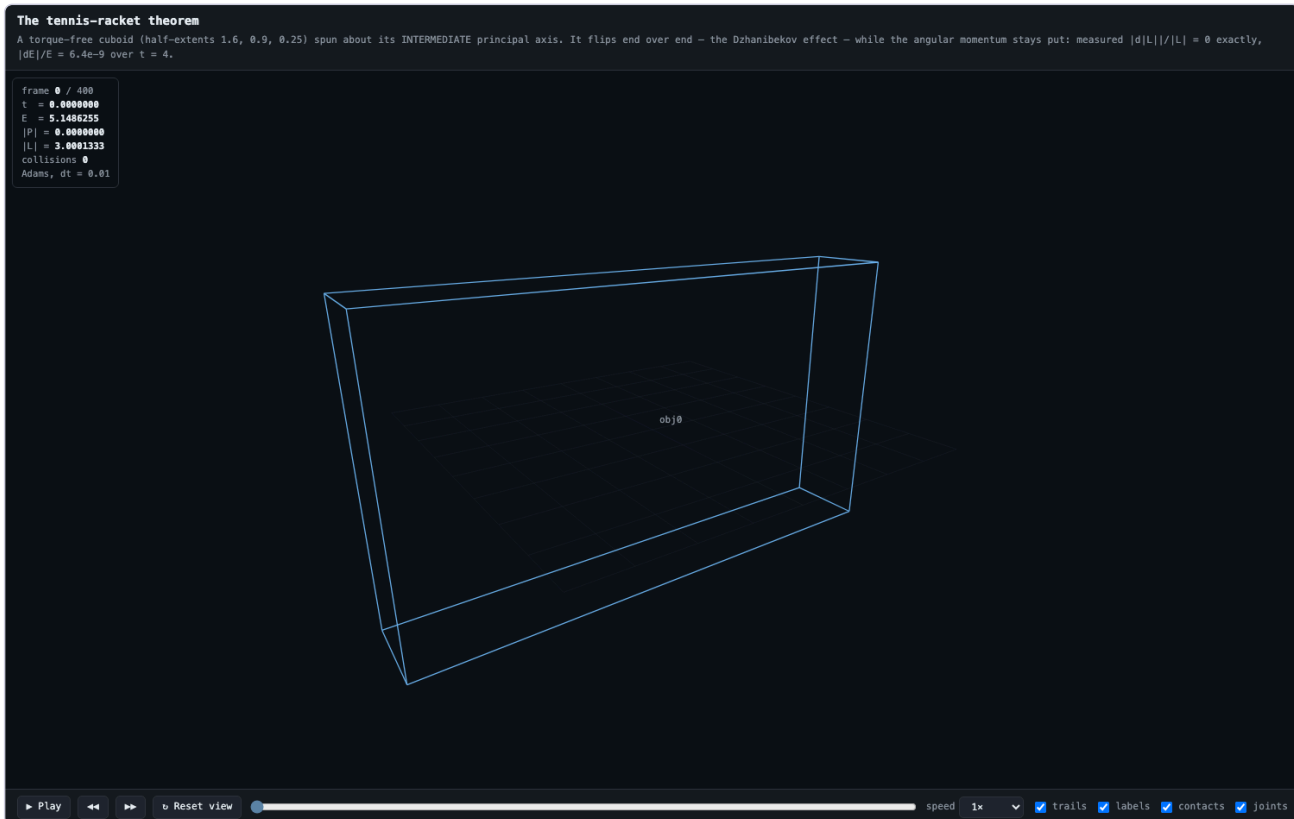


► Play ◀▶ ◂ Reset view speed 1x ☒ trails ☒ labels ☒ contacts ☒ joints

video_tumbling_racket

This notebook is one half of a pair. The other half is the example file `videos/scenes/tumbling_racket.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_tumbling_racket/video_tumbling_racket.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_tumbling_racket.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/tumbling_racket.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/tumbling_racket.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/tumbling_racket/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_tumbling_racket/player_video_tumbling_racket.py`
`run|gui|jump|capture out.png|data outdir`



video_universal_joint

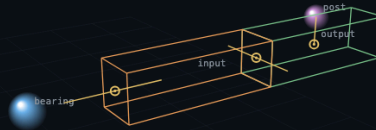
This notebook is one half of a pair. The other half is the example file `videos/scenes/universal_joint.posim` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (8 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/05_mechanism_videos/video_universal_joint/video_universal_joint.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/video_universal_joint.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/scenes/universal_joint.posim`
- **Movie page:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/videos/universal_joint.html`
- **Live GUI:** `python3 rustSolveIt_macos-silicon_SUNDIALS_7_8_0/gui/universal_joint/server.py`
- **Player:** `python3 SolveIt_Notebooks_for_rust/05_mechanism_videos/video_universal_joint/player_video_universal_joint.py`
`run|gui|jump|capture out.png|data outdir`

A universal joint transmitting rotation

A torque spins the input shaft up from rest; the Cardan joint carries the rotation across to the output shaft, which a rod braces to a post. The bend runs from straight to $\cos^{-1} 0.6 = 53.130$ degrees and back, and the speed ratio across the joint swings with it – a Cardan joint is not a constant-velocity joint.

```
frame 0 / 180  
t = 0.0000000  
E = 0.0000000  
|P| = 0.0000000  
|L| = 0.0000000  
collisions 0  
joints hinge, universal, rod  
worst |g| = 0.00e+0  
Ida, dt = 0.025
```



► Play



Reset view



speed

1x



☒ trails

☒ labels

☒ contacts

☒ joints

06_rust_compiled_examples

The compiled, self-checking Rust examples: each notebook builds and runs one `physical_object` example and asserts its `SUCCESS` verdict — the verdict is textual, so these entries carry no GUI image, honestly.

rust_bouncing_ball_restitution

This notebook is one half of a pair. The other half is the example file `physical_object/examples/bouncing_ball_restitution.rs` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (8 code cells)
- **Copy:**
`SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_bouncing_ball_restitution/rust_bouncing_ball_restitution.ipynb`
(header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/rust_bouncing_ball_restitution.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/physical_object/examples/bouncing_ball_restitution.rs`
- **Player:** python3
`SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_bouncing_ball_restitution/player_rust_bouncing_ball_restitution.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

rust_charged_in_b_field

This notebook is one half of a pair. The other half is the example file `physical_object/examples/charged_in_b_field.rs` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (10 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_charged_in_b_field/rust_charged_in_b_field.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/rust_charged_in_b_field.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/physical_object/examples/charged_in_b_field.rs`
- **Player:** python3
`SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_charged_in_b_field/player_rust_charged_in_b_field.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

rust_kepler_orbit

This notebook is one half of a pair. The other half is the example file `physical_object/examples/kepler_orbit.rs` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (10 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_kepler_orbit/rust_kepler_orbit.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/rust_kepler_orbit.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/physical_object/examples/kepler_orbit.rs`
- **Player:** python3 `SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_kepler_orbit/player_rust_kepler_orbit.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

rust_newtons_cradle

This notebook is one half of a pair. The other half is the example file `physical_object/examples/newtons_cradle.rs` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (11 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_newtons_cradle/rust_newtons_cradle.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/rust_newtons_cradle.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/physical_object/examples/newtons_cradle.rs`

- **Player:** `python3 SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_newtons_cradle/player_rust_newtons_cradle.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

`rust_outer_solar_system`

This notebook is one half of a pair. The other half is the example file `physical_object/examples/outer_solar_system.rs` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (11 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_outer_solar_system/rust_outer_solar_system.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/rust_outer_solar_system.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/physical_object/examples/outer_solar_system.rs`
- **Player:** `python3`
`SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_outer_solar_system/player_rust_outer_solar_system.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

`rust_tumbling_body`

This notebook is one half of a pair. The other half is the example file `physical_object/examples/tumbling_body.rs` in this repository, which contains the same simulation written directly in the simulator's own command language. This notebook runs that same simulation from Python, and explains every line of it.

- **Executed:** PASS — every cell green (9 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_tumbling_body/rust_tumbling_body.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/notebooks/rust_tumbling_body.ipynb`
- **Paired scene/source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/physical_object/examples/tumbling_body.rs`
- **Player:** `python3` `SolveIt_Notebooks_for_rust/06_rust_compiled_examples/rust_tumbling_body/player_rust_tumbling_body.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

07_nbody_rebound_rust

The N-body verification notebooks of the rebound/reboundx pure-Rust port: integrator equivalence, simulation archives, tides and spin, shearing sheets. Verdicts are textual (no browser GUI in this family).

addfmt_test

Adds the built-in solar system plus particles specified by orbital elements, by Pal coordinates, and by orbital period. Verified bit-identical against C.

- **Executed:** PASS — every cell green (3 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/addfmt_test/addfmt_test.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/addfmt_test.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/addfmt_test/player_addfmt_test.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

archive_test

Writes a 3-snapshot Simulationarchive. The same file loads in the C build and continues bit-identically, and vice versa — the formats are fully interchangeable.

- **Executed:** PASS — every cell green (3 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/archive_test/archive_test.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/archive_test.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/archive_test/player_archive_test.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

bs_pow_diff

The BS integrator picks its own step size using `pow()`. This evaluates exactly those `pow()` calls (200,000 samples) so the disagreement can be measured precisely: 56 differ, every one by exactly 1 ULP — the smallest difference two numbers can have.

[Encyclopedia note: the intro above describes the Windows/MSVC measurement; the executed run embedded in THIS copy reports zero `pow` divergences ('ULP distribution: none') on macOS/Apple Silicon.]

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/bs_pow_diff/bs_pow_diff.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/bs_pow_diff.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/bs_pow_diff/player_bs_pow_diff.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

derivatives_test

Evaluates all 65 `reb_particle_derivative_*` functions on two configurations and dumps raw bits; verified 130/130 lines bit-identical against the C build.

- **Executed:** PASS — every cell green (3 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/derivatives_test/derivatives_test.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/derivatives_test.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/derivatives_test/player_derivatives_test.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

frequency_test

Runs REBOUND's frequency analysis in all three modes on a synthetic three-frequency signal (true frequencies 0.30, 0.55, 0.11 radians per sample) and prints the recovered frequencies, amplitudes and phases.

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/frequency_test/frequency_test.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/frequency_test.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/frequency_test/player_frequency_test.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

integrators_test

Runs the fixed three-body problem under a chosen integrator configuration and dumps the final state as raw IEEE-754 bits. 63 configurations of this harness were verified bit-identical against the MSVC C build. Change the arguments below to try others (e.g. 'ias15', 'saba-h1064', 'mercurius', 'trace').

[Encyclopedia note: the bit-identical-vs-MSVC claim is history from the port's Windows lineage; the verdict embedded in THIS copy is the macOS/Apple Silicon run's own.]

- **Executed:** PASS — every cell green (3 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/integrators_test/integrators_test.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/integrators_test.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/integrators_test/player_integrators_test.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

kepler_rectilinear

Calls the Kepler solver with (near-)rectilinear hyperbolic motion — the regime that strains its iteration hardest. This probe found a genuine port defect: with a large timestep the Rust looped forever where the C returned, because while ($a > b$) and if ($a \leq b$) break differ when a value is NaN. Now fixed; the two agree bit-for-bit.

- **Executed:** PASS — every cell green (3 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/kepler_rectilinear/kepler_rectilinear.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/kepler_rectilinear.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/kepler_rectilinear/player_kepler_rectilinear.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

libm_diff

Samples 200,000 inputs per maths function and dumps raw bits, so the C and Rust results can be compared exactly. This is the measurement that underpins the whole project: sin, cos, tan, atan2, sqrt, fmod, exp, log and cbrt are bit-identical to Microsoft's C library, and pow is the one exception.

[Encyclopedia note: the intro above is the notebook's own Windows-lineage description; the executed run embedded in THIS copy reports 'functions that differ: NONE - all bit-identical' on macOS/Apple Silicon.]

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/libm_diff/libm_diff.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/libm_diff.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/libm_diff/player_libm_diff.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

movetocom_var

Probes `reb_simulation_move_to_com` with MEGNO variational particles. An audit found the first-order shift summed the wrong particle array, which silently changed every MEGNO/Lyapunov result. Now fixed and verified bit-identical against C.

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/movetocom_var/movetocom_var.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/movetocom_var.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/movetocom_var/player_movetocom_var.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

movetocom_var_test

The probe written during the audit to demonstrate the variational centre-of-mass defect, kept as a regression check. It prints both dm accumulators so you can see why the wrong array mattered.

- **Executed:** PASS — every cell green (3 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/movetocom_var_test/movetocom_var_test.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/movetocom_var_test.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/movetocom_var_test/player_movetocom_var_test.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

rebx_binary_roundtrip

Serialises a REBOUNDx state (forces and parameters of several types), reads it back into a fresh simulation, and checks every value returns with identical bits.

- **Executed:** PASS — every cell green (3 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/rebx_binary_roundtrip/rebx_binary_roundtrip.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/rebx_binary_roundtrip.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/rebx_binary_roundtrip/player_rebx_binary_roundtrip.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

server_test

Starts the ported web server, fetches the /simulation binary over HTTP, then shuts it down with the 'Q' key endpoint. The served blob is a valid REBOUND binary that the C build loads to the identical state.

- **Executed:** PASS — every cell green (2 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/server_test/server_test.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/server_test.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/server_test/player_server_test.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

shearing_sheet

A straight port of REBOUND's flagship example: a small sheared box of colliding ice particles, using the SEI integrator, octree self-gravity, tree collision search and shear-periodic boundaries. Note: like the stock C example, shearing_sheet integrates to infinity — you stop it with Ctrl+C. A notebook cannot run that, so this notebook builds it (to prove it compiles) and then runs the terminating seeded harness shearing_sheet_test for 400 steps to produce the plot. The two set up identical physics; the harness just fixes the random seed and stops.

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/shearing_sheet/shearing_sheet.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/shearing_sheet.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/shearing_sheet/player_shearing_sheet.py run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

shearing_sheet_test

Runs the seeded 400-step shearing sheet and, if the C reference dump is present, verifies the two agree on every bit and prints both SHA-256 fingerprints. This is the project's headline acceptance test: 1,482 particles and ~102,500 collisions.

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/shearing_sheet_test/shearing_sheet_test.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/shearing_sheet_test.ipynb`

- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/shearing_sheet_test/player_shearing_sheet_test.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

tides_spin_kozai

A planet with a distant stellar companion that periodically drives its orbit to high eccentricity. Uses the ADAPTIVE IAS15 integrator plus two REBOUNDx forces at once. Matching the C bit-for-bit here means both programs chose the identical sequence of thousands of adaptive steps.

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/tides_spin_kozai/tides_spin_kozai.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/tides_spin_kozai.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/tides_spin_kozai/player_tides_spin_kozai.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

tides_spin_migration

Two Earth-sized planets migrating through a gas disk while tides evolve their spins. Exercises two REBOUNDx forces simultaneously and a parameter changed in the middle of the run (migration is switched off half-way). Verified bit-identical to the C.

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/tides_spin_migration/tides_spin_migration.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/tides_spin_migration.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/tides_spin_migration/player_tides_spin_migration.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

tides_spin_pseudo

A giant planet orbiting very close to its star, starting slightly elliptical, tilted 30 degrees and spinning fast. Tides should circularise the orbit, damp the tilt, and settle the spin. Exercises WHFast plus REBOUNDx's tides_spin force and its spin differential equations. Verified bit-identical to the C REBOUNDx.

- **Executed:** PASS — every cell green (4 code cells)
- **Copy:** `SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/tides_spin_pseudo/tides_spin_pseudo.ipynb` (header cell prepended: run commands, GUI commands, movies, data access, full player script)
- **Canonical source:** `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/rebound_rust/notebooks/tides_spin_pseudo.ipynb`
- **Player:** `python3 SolveIt_Notebooks_for_rust/07_nbody_rebound_rust/tides_spin_pseudo/player_tides_spin_pseudo.py`
`run|gui|jump|capture out.png|data outdir`

(no browser GUI in this family — the verdict is textual, embedded in the executed notebook)

Appendix A — excluded duplicates

The workspace holds several parallel copies of the same notebooks; the encyclopedia uses one canonical edition of each and excludes, deliberately: the 109 earlier editions under `rustSolveIt_Using_SUNDIALS_7_8_0/version-7.8.0/notebooks/` (the pre-macOS lineage of the same notebooks), the two engine-mirror copies of the planet-Mercury notebooks under `rustSolveIt_macos-silicon_SUNDIALS_7_8_0/planet_Mercury/notebook/`, two strays at the workspace root (one byte-identical to its canonical copy; the other an earlier edition of the test-2 notebook superseded by the canonical one), and the duplicated `docs/ipython_examples` copy inside the vendored rebound sources. `StageNbks/*.posim` are posim REPL sessions, not Jupyter notebooks, and are documented in the engine's own `STAGENBKS_PROVENANCE.md`.

Appendix B — vendored upstream reference notebooks (not executed)

The vendored third-party rebound / REBOUNDx sources ship their own example notebooks (they exercise the upstream C libraries, not this project's Rust engine, and so are catalogued here as reference only):

- AdvWHFast.ipynb
- CentralForce.ipynb
- ChaoticHyperion.ipynb
- Cheartbeat.ipynb
- Checkpoints.ipynb
- Churyumov-Gerasimenko.ipynb
- CloseEncounters.ipynb
- CustomSplittingIntegrationSchemes.ipynb
- Custom_Effects.ipynb
- EccAndIncDamping.ipynb
- EccentricComets.ipynb
- EmbeddedOperatorSplittingMethods.ipynb
- EscapingParticles.ipynb
- ExponentialMigration.ipynb
- Forces.ipynb
- FourierSpectrum.ipynb
- FrequencyAnalysis.ipynb
- GasDampingTimescale.ipynb
- GasDynamicalFriction.ipynb
- GeneralRelativity.ipynb
- GettingStartedParameters.ipynb
- HighOrderSymplectic.ipynb
- Holmberg.ipynb
- Horizons.ipynb
- HybridIntegrationsWithTRACE.ipynb
- HyperbolicOrbits.ipynb
- InnerDiskEdge.ipynb
- IntegrateForce.ipynb
- IntegratingArbitraryODEs.ipynb
- J2.ipynb
- LenseThirring.ipynb
- Megno.ipynb
- Migration.ipynb
- ModifyMass.ipynb
- ObliquityEvolution.ipynb
- OperatorsOverview.ipynb
- OrbitPlot.ipynb
- OrbitalElements.ipynb
- ParameterInterpolation.ipynb
- PoincareMap.ipynb
- PoincareSurfaceOfSection.ipynb
- PrimordialEarth.ipynb
- RadialVelocity.ipynb
- Radiation_Forces_Circumplanetary_Dust.ipynb
- Radiation_Forces_Debris_Disk.ipynb
- RealtimeVisualizations.ipynb
- Resonances_of_Jupiters_moons.ipynb
- Rotations.ipynb
- SaturnsRings.ipynb
- SavingAndLoadingSimulations.ipynb
- Simulationarchive.ipynb
- SimulationarchiveRestart.ipynb
- SpinsIntro.ipynb
- Starman.ipynb
- StochasticForces.ipynb

- [StochasticForcesCartesian.ipynb](#)
- [Testparticles.ipynb](#)
- [TidesConstantTimeLag.ipynb](#)
- [TidesDynamical.ipynb](#)
- [TidesSpinEarthMoon.ipynb](#)
- [TidesSpinPseudoSynchronization.ipynb](#)
- [TrackMinDistance.ipynb](#)
- [TransitTimingVariations.ipynb](#)
- [TypeIMigration.ipynb](#)
- [UniquelyIdentifyingParticlesWithNames.ipynb](#)
- [Units.ipynb](#)
- [User_Defined_Collision_Resolve.ipynb](#)
- [VariationalEquations.ipynb](#)
- [VariationalEquationsWithChainRule.ipynb](#)
- [WHFast.ipynb](#)
- [WHFast512.ipynb](#)
- [YarkovskyEffect.ipynb](#)

Appendix C — reproduction commands

```
# from the workspace root:
cd rustSolveIt_macos-silicon_SUNDIALS_7_8_0 && cargo build --release -p posim && cd ..
cd planet_Mercury/mercury_rs && cargo build --release && cd ../../..
python3 SolveIt_Notebooks_for_rust/_tools/gen.py      # regenerate copies+players
python3 SolveIt_Notebooks_for_rust/_tools/run_copy.py SolveIt_Notebooks_for_rust/<topic>/<tag>/<tag>.ipynb
python3 SolveIt_Notebooks_for_rust/_tools/shoot.py    # re-capture every GUI image
python3 SolveIt_Notebooks_for_rust/_tools/encyclo.py  # rebuild this encyclopedia
```